Satellite Communications
ECE 4644 / 5610 Spring 2026

Week #13

April 13 - 17, 2026

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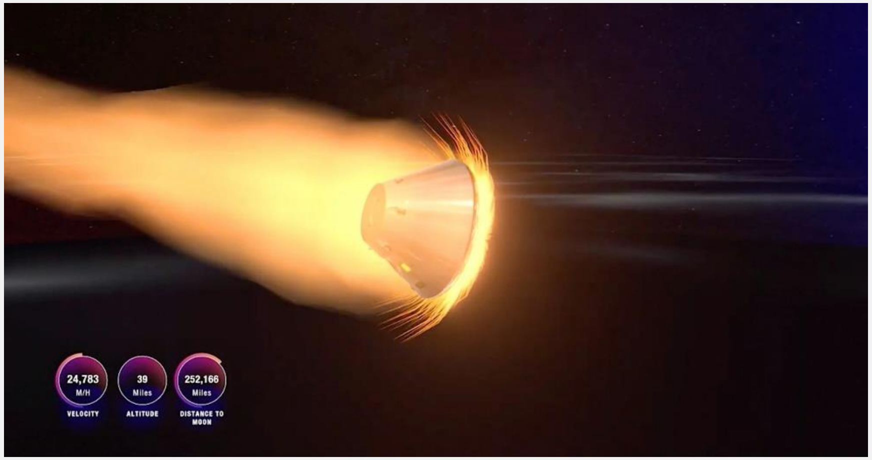
Topics

- Multiple Access (MA) techniques:
- FDMA
- TDMA
- CDMA
- RA
- Demand Assignment / Demand Assignment Multiple Access (DAMA)

Artemis II Update: Splashdown on April 10, 8:12 PM

Artemis II – Radio blackout on descent

- 7:53 PM: Nearing 120 km altitude the Orion s/c experienced a planned communications blackout (~ 6 min)
- The outage is due to the build-up of plasma as the s/c performs a fiery re-entry and ionizes the atmosphere
- In a sufficiently dense plasma, RF signal does not propagate



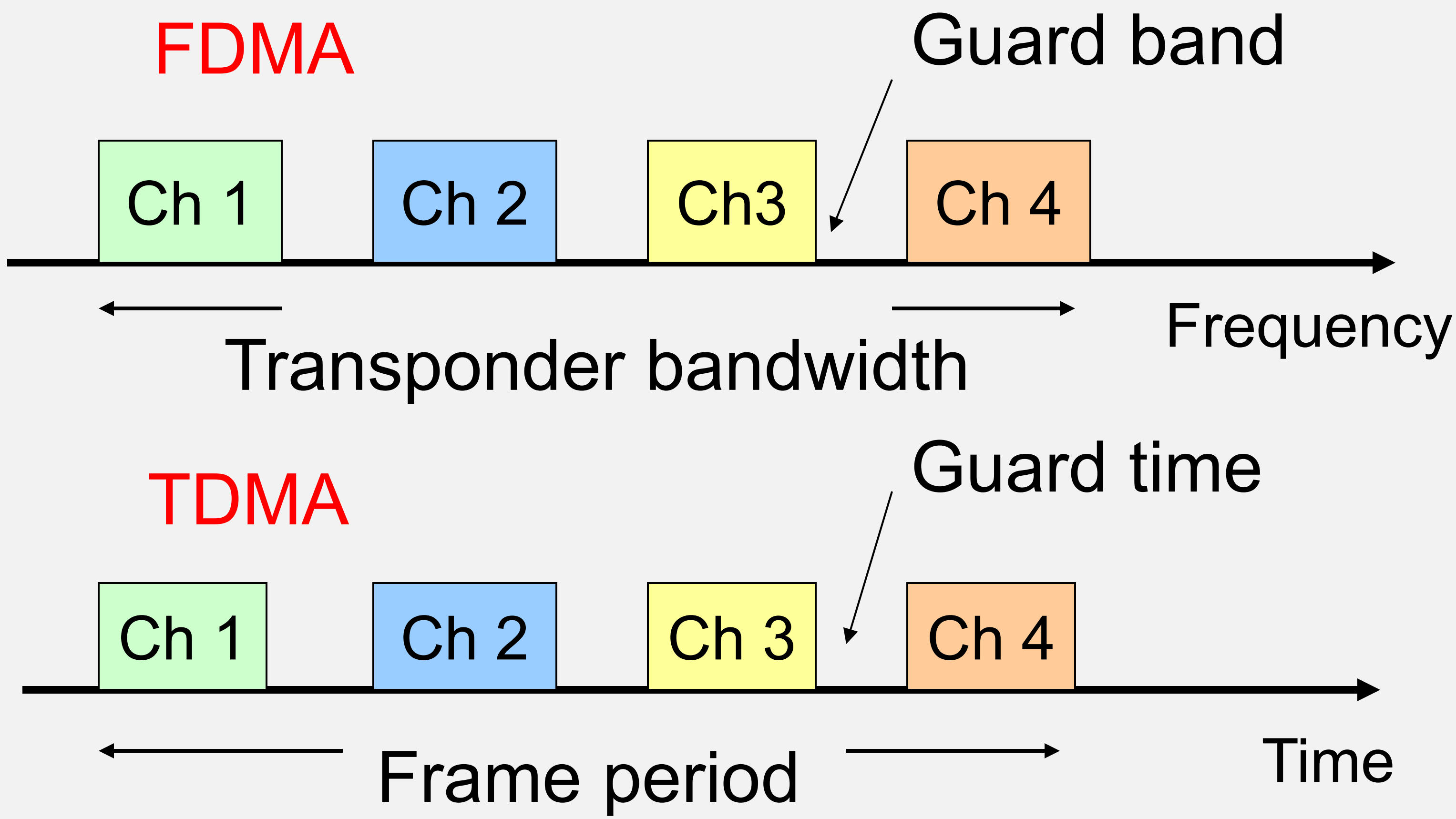
Multiple Access Techniques - Review

- Communication satellites carry many signals at the same time using *Multiple Access* (MA) techniques
- These make use of frequency, time, or code to label individual signals in the presence of other signals
- The choice of MA technique is a design consideration for a Satcom system

MA Techniques

- FDMA (Frequency Division Multiple Access)
  - Signals are separated in frequency
- TDMA (Time Division Multiple Access)
  - Signals are separated in time
- CDMA (Code Division Multiple Access)
  - Signals are separated by coding
- Random Access (RA)
  - Signals are transmitted independently of one another

Fig. 6.9


FDMA
Guard band
Ch 1
Ch 2
Ch3
Ch 4
Frequency
Transponder bandwidth
TDMA
Guard time
Ch 1
Ch 2
Ch 3
Ch 4
Time
Frame period

FDMA / TDMA / CDMA / RA

- With FDMA, signals are assigned to distinct bands within a designated transponder bandwidth
- With TDMA, signals occupy distinct time slots
- With CDMA, transmission occurs at the same frequency and time; coding is used to extract individual signals
- With RA, there is no effort to separate the signals

FDMA - Tx

- FDMA is widely used as a way to share transponder bandwidth using multiple carrier frequencies and guard bands
- Transponder bandwidths of 36, 54, and 72 MHz are typical
- Earth stations are assigned to channels within the bandwidth

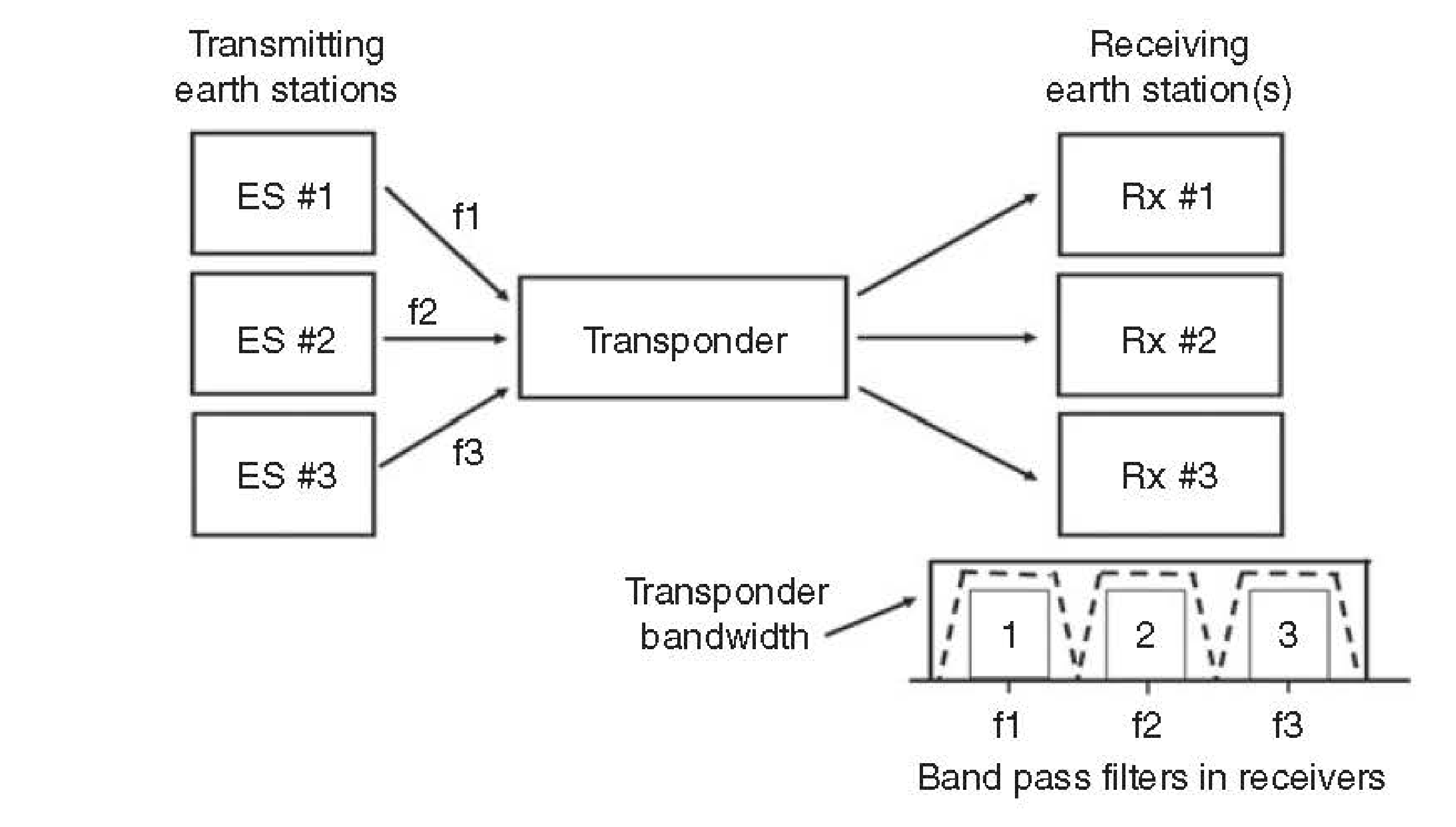
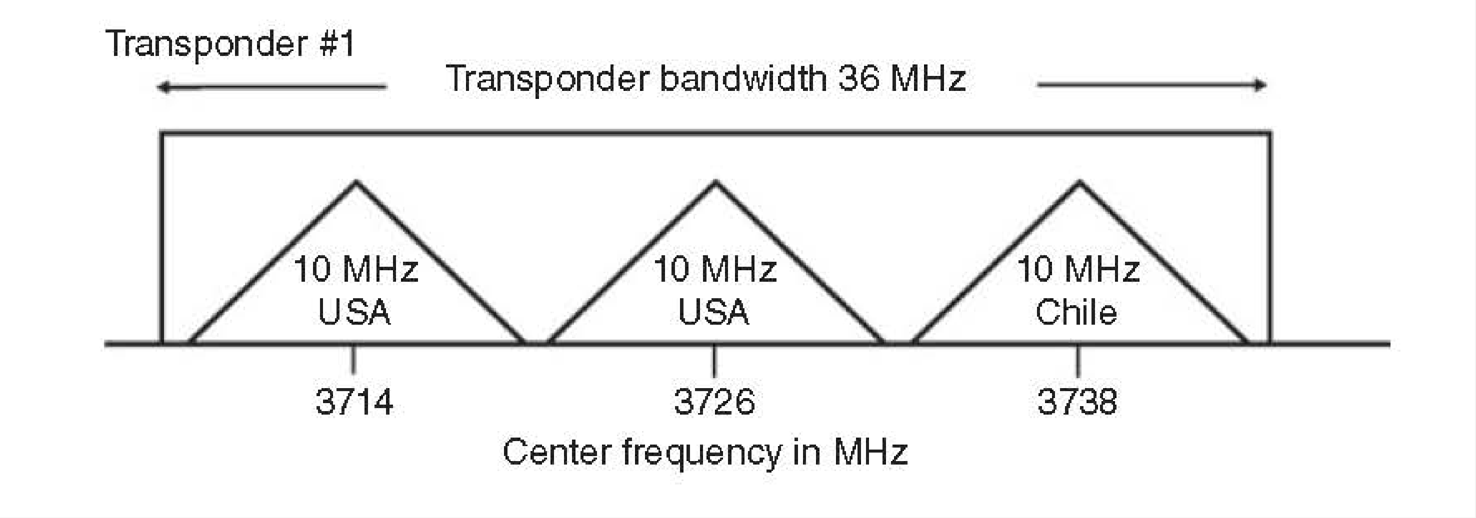


Illustration of FDMA (Figure 6.11 from course text)

FDMA - Tx

- For Tx the signals are mapped at the IF stage to distinct frequency bands
- These upconvert to a mapping of frequency bands about the RF carrier frequency
- The frequency bands fit within the transponder bandwidth with guard bands applied



FDMA plan for a C-band transponder
(from Figure 6.12 from course text)

FDMA - Rx

- For Rx the Earth station downconverts all the signals to IF
- Selects for the desired signal frequency with a narrowband filter
- A large numbers of IF receivers is needed if the station wants to be able to receive all the signals
- Observe that the noise situation for FDMA is determined by the narrowband Rx filter

FDMA – Intermodulation Products

- Interference between signals in the transponder gives rise to Intermodulation products (IM), a type of noise that can be associated with a $(C/N)_{IM}$
- Encountered when transponder operates near saturation (high power), in the nonlinear regime
- IM is countered by applying backoff to the transponder power, which lowers $(C/N)_{dn}$
- Systems are designed to find the 'sweet spot' in terms of trading off $(C/N)_{dn}$ and $(C/N)_{IM}$ to maximize $(C/N)_o$

FDM versus FDMA

- Do not confuse *Frequency Division Multiplexing* (FDM) with FDMA
- FDM refers to signals being multiplexed onto a transmission link at a single Earth source
- FDMA has multiple Earth sources of signal
- FDM can be combined with FDMA to produce *FDM – FDMA*
- Many such hybrid modes are possible

TDMA

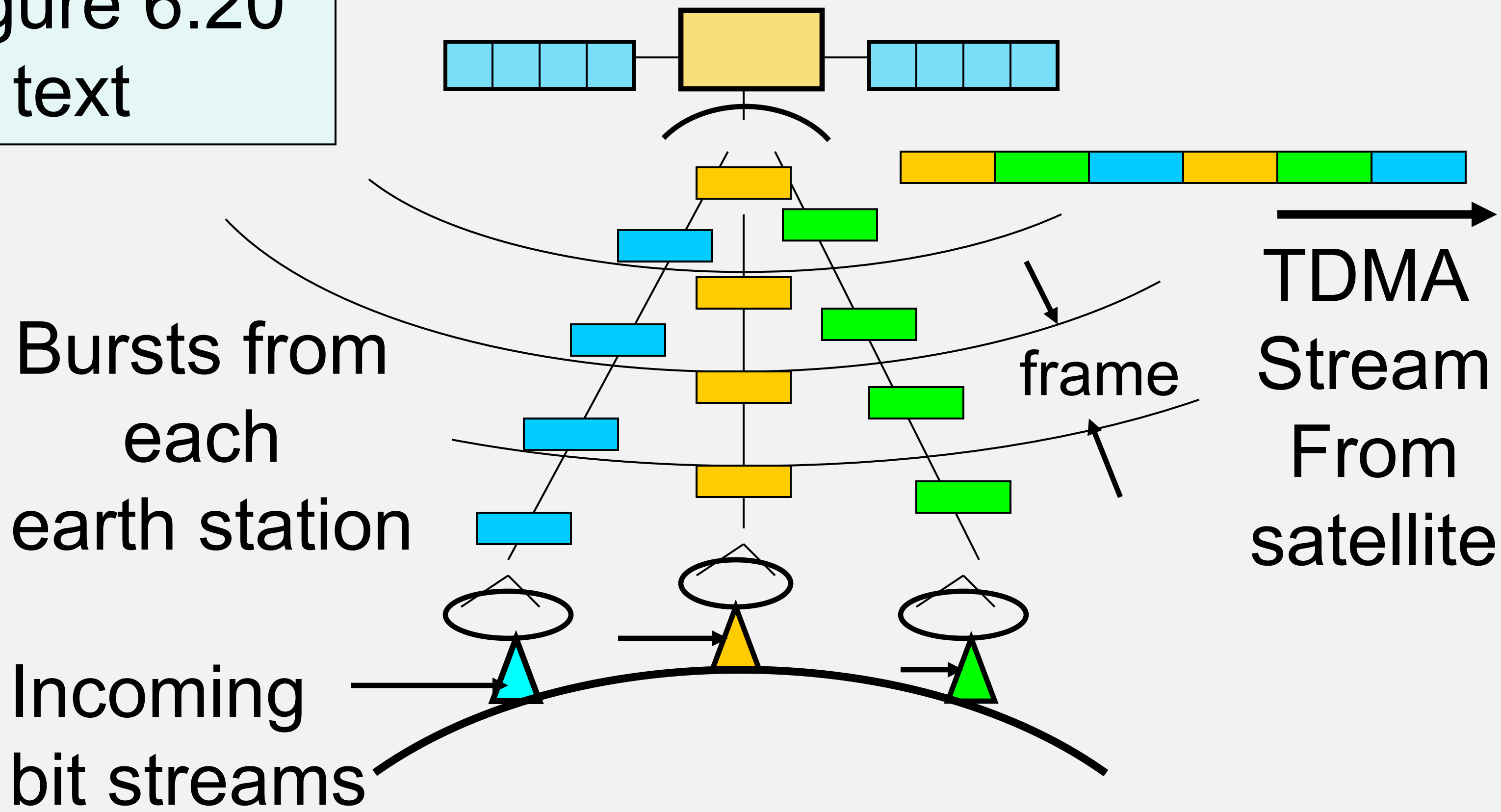
- Signals are separated in time
  - Transponder bandwidth is common to all signals
  - Stations transmit *bursts* of signal in sequence
- Each earth station transmits *in sequence*
- Transmitted bursts from many earth stations arrive at the satellite

  - in an orderly fashion

  - in the correct order

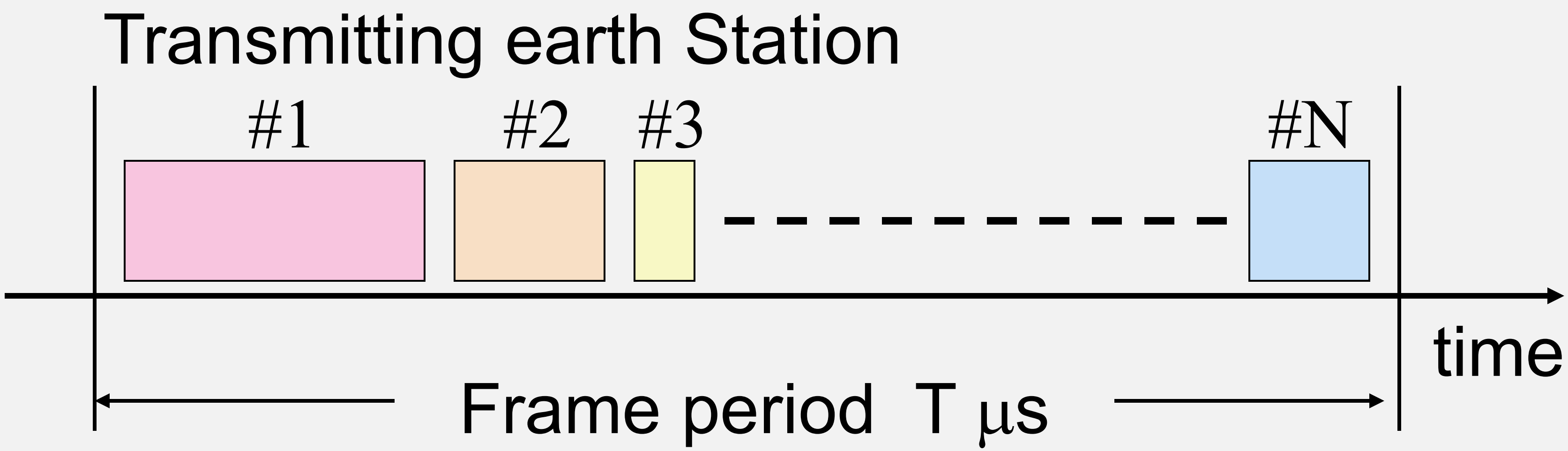
TDMA Basics

Figure 6.20 in text



TDMA Frame Structure

From Figure 6.21 in text



Users occupy a specific time within the frame

Guard times are required between bursts

TDMA Frames

- Frames consist of sequence of bursts from each earth station
- Bursts are completely independent QPSK or QAM transmissions
- Bursts must be separated by guard times to prevent overlap (called *collisions*)
- Some form of start of frame marker is required - could be from control station, or start of station #1 burst

TDMA Frames

- Transmissions are divided into a preamble and traffic bits
- Preamble carries information necessary to identify the bursts and extract their content
- Traffic bits earn revenue while preamble represents overhead

TDMA versus TDM

- Do not confuse TDMA with Time Division Multiplexing (TDM)
- TDM refers to signals being multiplexed in time onto a transmission link at a single location
- TDMA refers to a single transponder being accessed by RF carriers from different Earth stations
- Systems can be structured as TDM-TDMA

TDMAs and Digital Speech

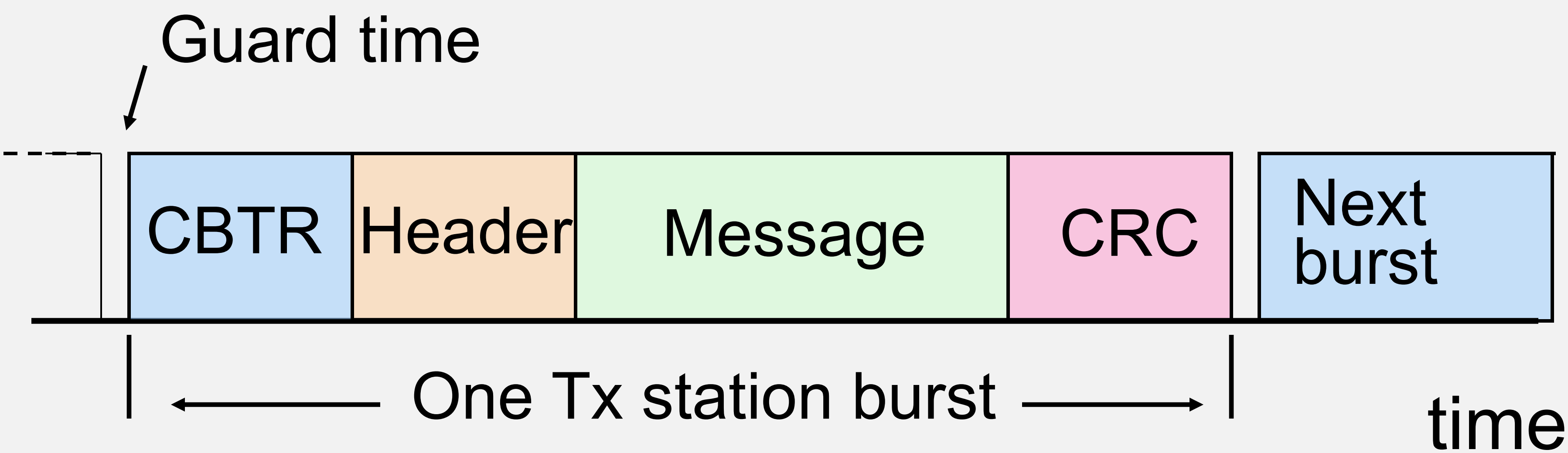
- Early TDMAs were designed with 125 μs frames = 8 kHz repetition rate
- Matches the 8 kHz sampling rate of digital voice as 8-bit digital words
- So a frame transmits one word from each voice channel
- This is sufficient to maintain each voice channel connection in real time

TDMAs and Digital Speech

- The TDMA frame can be lengthened to 2 ms (which is 16x greater)
- Leading to greater efficiency from improved message / preamble ratio
- Now each voice channel will need to deliver sixteen 8-bit words within a frame

TDMA Burst Structure

- Burst looks like a packet with added CTBR



CBTR = carrier and bit timing recovery
Header contains addresses, etc
CRC is cyclic redundancy check

TDMA CBTR

- Each received burst in a TDMA frame is independent
- QPSK demod must lock onto carrier of each burst (carrier recovery to enable bit recovery)
- CBTR is unmodulated carrier + 10101 pattern (Unique Word)
- 176 symbols typical length (3 μs at 60 Msps)
- Demod must synchronize in < 3 μs
- Bit clock must synchronize in < 3 μs

Efficiency in TDMA

- Headers, CBTR, CRC, guard times all waste time that could be used to send data
- Longer frame sends more data, less *overhead*, but adds latency
- Typical TDMA frames are 2 - 20 ms long
- Message bit rate = $R_b \times N / (N + M)$
- where N = number of message bits in frame

  M = total possible bits in frame

Efficiency in TDMA - Example

- Frame length 2.0 ms
- QPSK with symbol rate 50 Msps
- Overhead is 1000 symbols / burst
- Five earth stations with 2 μs guard times
- What is the efficiency of data transfer?

Efficiency in TDMA - Example

- Symbols per frame = $2\ 10^{-3}$ ms x $50\ 10^{6}\ s^{-1}$ = 100,000 symbols
- Overhead per frame = 5 x 1000 = 5,000 symbols
- Guard time per frame = 5 x $2\ 10^{-6}$ ms x $50\ 10^{6}\ s^{-1}$ = 500 symbols
- Lost symbols per frame = 5,500
- Data symbols per frame = 94,500
- Efficiency = 94,500 / 100,000 = 94.5%

TDMA Considerations

- Observe that each burst occupies the full bandwidth of the transponder by itself
- So, less problem with signals interfering in the transponder, i.e., no IM products, less backoff needed

TDMA Considerations

- <u>But</u> the signal is comparatively wideband
- Earth station must transmit at a high bit rate (to occupy full transponder bandwidth)
- Need to operate with higher transmitter power compared to FDMA to obtain same C/N

When to use TDMA

- TDMA is good for large fixed networks, also used in some mobile networks
- Not as good for VSATs or satphones (power limited)
- Requires a high bit rate compared to FDMA:

  To transmit M stations:

  Receiver b/w = M x Channel b/w for FDMA
- C/N in receiver is reduced by a factor of M due to increase in noise power

When not to use TDMA - example

- Compare a 100-channel system using FDMA at 5 ksps to one using TDMA bursts at 500 ksps
- FDMA: CNR = + 13.0 dB
- For TDMA C/N in receiver is 1 / M x FDMA case
- TDMA: CNR = - 7.0 dB
- Attributable to increase in noise power due to 100x increase in R_s and B_{occ}

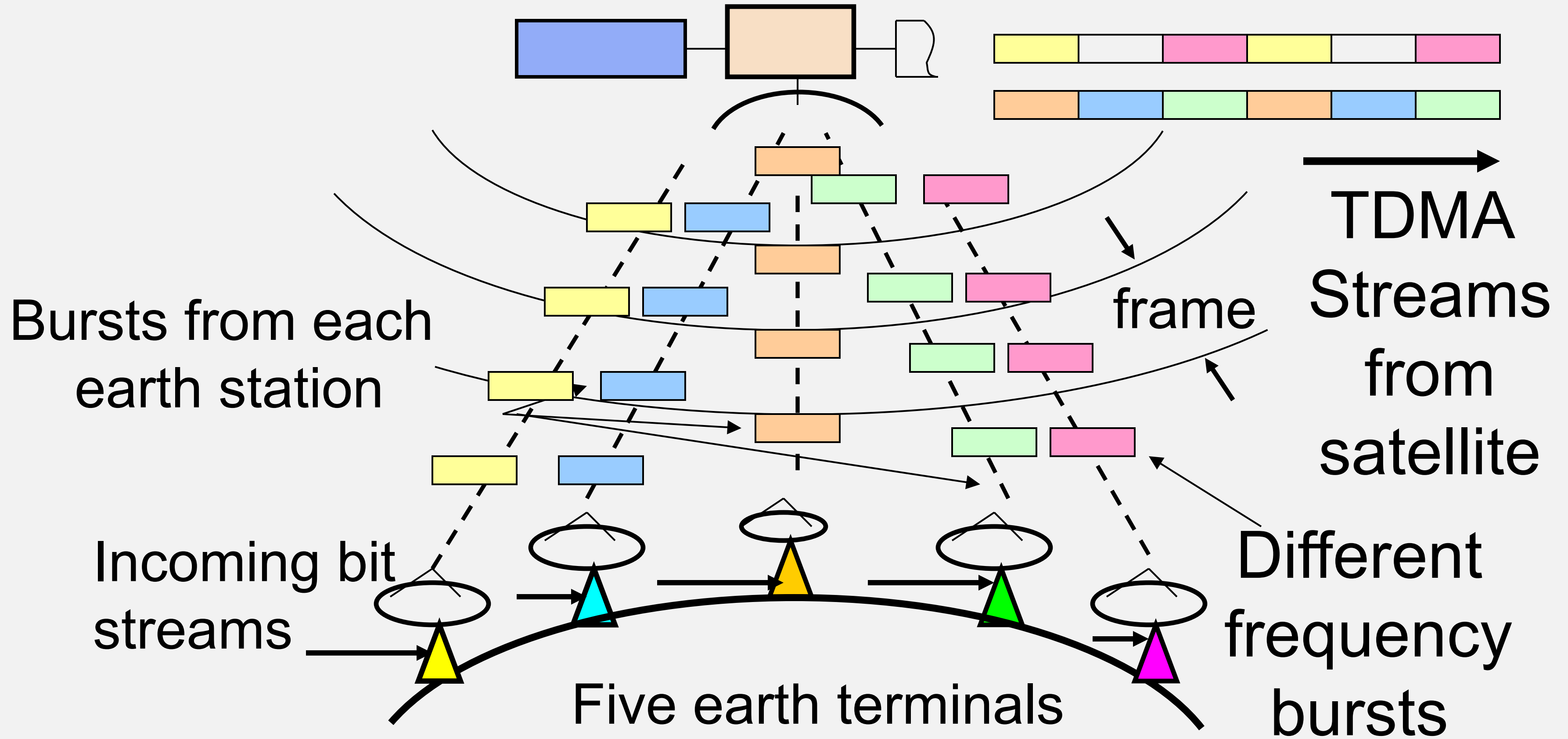
When not to use TDMA - example

- To compensate we need to boost the transmit power or EIRP by a corresponding amount
- Moral : TDMA can be used only where there is sufficient EIRP in transmitter (because of high bit rate requirement)
- So, TDMA is not suitable for power-limited stations (VSATs, satphones)

MF - TDMA

- TDMA can be used with FDMA to make MF-TDMA (Multi-frequency TDMA)
- Used in some VSAT networks
- Small number of stations form TDMA frame
- Many such groups share transponder in FDMA
- Reduces bandwidth requirement compared to full TDMA but now some IM products

MF – TDMA Internet Access



TDMA timing issues

- Problem
  - Delay time to GEO satellite is ~120 ms
  - TDMA Frame length is 125 μs to 2 ms
  - There could be almost 1000 frames on the path to the satellite!
- Accurate timing is very important
- Earth stations must stay in synch

TDMA Timing

- Timing obtained by:
  - Organizing TDMA transmission into frames
  - Transmitting bursts of signals from earth stations
  - Synchronizing all transmissions
- Minimum frame length is 125 μs
- 125 μs ≡ one voice channel

TDMA example: Voice

- A voice channel is also called *a terrestrial channel*
- Speech is digitized into 8-bit words
- Sampling rate of terrestrial channel is 8 kHz corresponding to one word every 125 μs
- The data rate for a terrestrial channel is therefore 8 bits x 8 kHz = 64 kbit/s

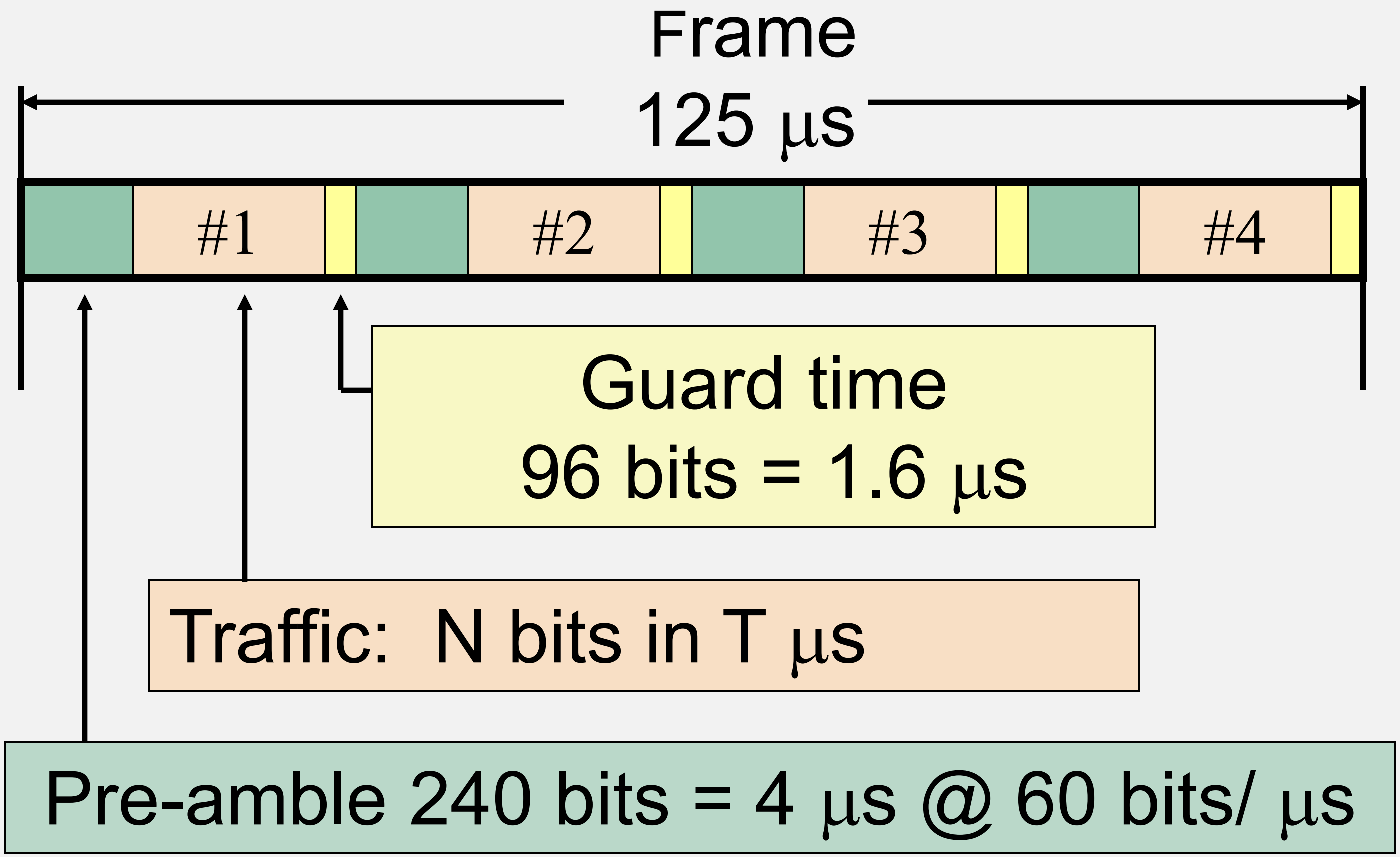
TDMA frames

- To reduce overhead, use longer frames
  - 125 μs frame: 1 Word/Frame
  - 500 μs frame: 4 Words/Frame
  - 2000 μs frame: 16 Words/Frame

TDMA example: Short frame

- Transponder bandwidth = 36 MHz
- Bit rate (QPSK) 60 Mbit/s = 60 bits/μs
- Four earth stations share transponder in TDMA using 125 μs frames
- Each station handles multiple voice channels
- Preamble = 4 μs
  = 240 bits
- Guard time = 1.6 μs
  = 96 bits

TDMA example: Short frame



TDMA example

- For this TDMA example
  - (a) What is the transponder capacity in terms of 64 kbit/s speech channels?
  - (b) How many channels can each earth station transmit?
- **Answer**
  - (a) There are four earth stations transmitting within the 125 μs frame …

TDMA example: Short frame

4 μs pre-amble, 1.6 μs guard time, T μs traffic bits

- 125 μs frame gives

$T_{\mathbf{frame}} = 125\ \mu s = (4 \times 4\mu s) + (4 \times 1.6\mu s) + (4 \times T_{\mathbf{traffic}}\ \mu s)$

Hence

$T_{\mathbf{traffic}} = (125 - 16 - 6.4) / 4 = 25.65\ \mu s$

60 Mbit/s $\equiv$ 60 bits/μs

25.65 μs = 1539 bits

Channels / earth station =1539 / 8 = 192 (.375)

TDMA example: Short frame

– (a) How many channels can each earth station transmit?

Answer: 192 (64 kbit/s) voice channels

– (a) What is the transponder capacity in terms of 64 kbit/s speech channels?

Answer: 4 x 192 = 768 (64 kbit/s) voice channels

TDMA example: Long Frame

- Now lengthen TDMA frame to 2 ms
- One voice channel contains sixteen 8-bit words
- $T_{\textbf{frame}} = 2 \text{ ms}$

  $= 2{,}000\ \mu\text{s} = 4 \times 4 + 4 \times 1.6 + 4 \times T_{\textbf{traffic}}$

  $T_{\textbf{traffic}} = 494.4\ \mu\text{s}$

  $T_{\textbf{traffic}} \Rightarrow 29{,}664$ bits (60 bits / μs)
- With 16 x 8 = 128 bits for a channel

  Channels / station = 29,664 / 128 = 231 (.75)

TDMA examples: Long versus short Frame

- Compare 2 ms frame versus 125 μs frame:
- Capacity has increased - lower overhead
- 125 μs frame $\Rightarrow$ 192 channels per station

  2 ms frame $\Rightarrow$ 231 channels per station

TDMA Synchronization

- Start-up requires care! See Section 6.6
- Master station generates a reference burst to mark the <u>s</u>tart <u>o</u>f a <u>t</u>ransmit <u>f</u>rame (SOTF)
- Stations adjust their burst timing using this master reference
- Need to have accurate range to satellite
- Clocks can also be locked to GPS time

TDMA and small capacity terminals

- TDMA is not a good multiple access choice for low capacity systems
- Example: Ku-band VSAT terminal
- Try FDMA with QPSK modulation, half rate FEC

R_b = 128 kbps for message data

R_b = 256 kbps after FEC is applied

R_s = 128 ksps

TDMA and small capacity terminals

- Receiver has B_n = 128 kHz
- Typically:

  P_t = 1 W

  G_t = 41 dB with 1 m dish at 12 GHz
- EIRP = 41 dBW for this FDMA plan

TDMA and small capacity terminals

- Now try TDMA with five earth stations in each frame
- Terminal must transmit bursts at

  R_s = 5 x 128 ksps = 640 ksps
- Receiver must have B_n = 640 kHz

  7 dB increase in receiver bandwidth

  Noise power in receiver increases by 7 dB

TDMA and small capacity terminals

- We must transmit

  EIRP = 41 + 7 = 48 dBW

  to achieve same BER and same margin
- Increase P_t to 5 W (exceeds regulated limit – stop!)
- Increase G to 48 dB - requires 2.24 m diameter antenna
- FDMA achieves same performance with 1 m dish

TDMA Summary

- Advantages
  - No intermodulation products (if the full transponder b/w is occupied)
  - Saturated transponder operation possible
  - Good for data efficiency
  - TDMA can optimize capacity per connection by using flexible Burst Time Plan
  - Useful with satellite switched beams

TDMA Summary

- Disadvantages
  - Complex
  - Every earth station *must* transmit at the same burst rate, irrespective of traffic
  - Demands higher EIRP than FDMA
  - High burst rate = more power per transmitter
  - Earth stations must stay in synch
  - Difficult to implement for intermittent users

Code Division Multiple Access - *CDMA*

- All users use the transponder bandwidth at the same time
- Each transmitter is allocated a unique code
- Signals are identified at a receiver by their unique codes
- Codes must be *orthogonal* to avoid interference: station A does not respond to code B

CDMA Codes

- CDMA codes are 16 bits to many thousands of bits in length
- The CDMA code bits are called *chips*
- Each message bit is encoded using the entire sequence of chips for the code
- The chip rate is much higher than the data rate (obviously), increasing the bandwidth
- CDMA is also known as *spread spectrum*

Direct Sequence Spread Spectrum (CDMA) System

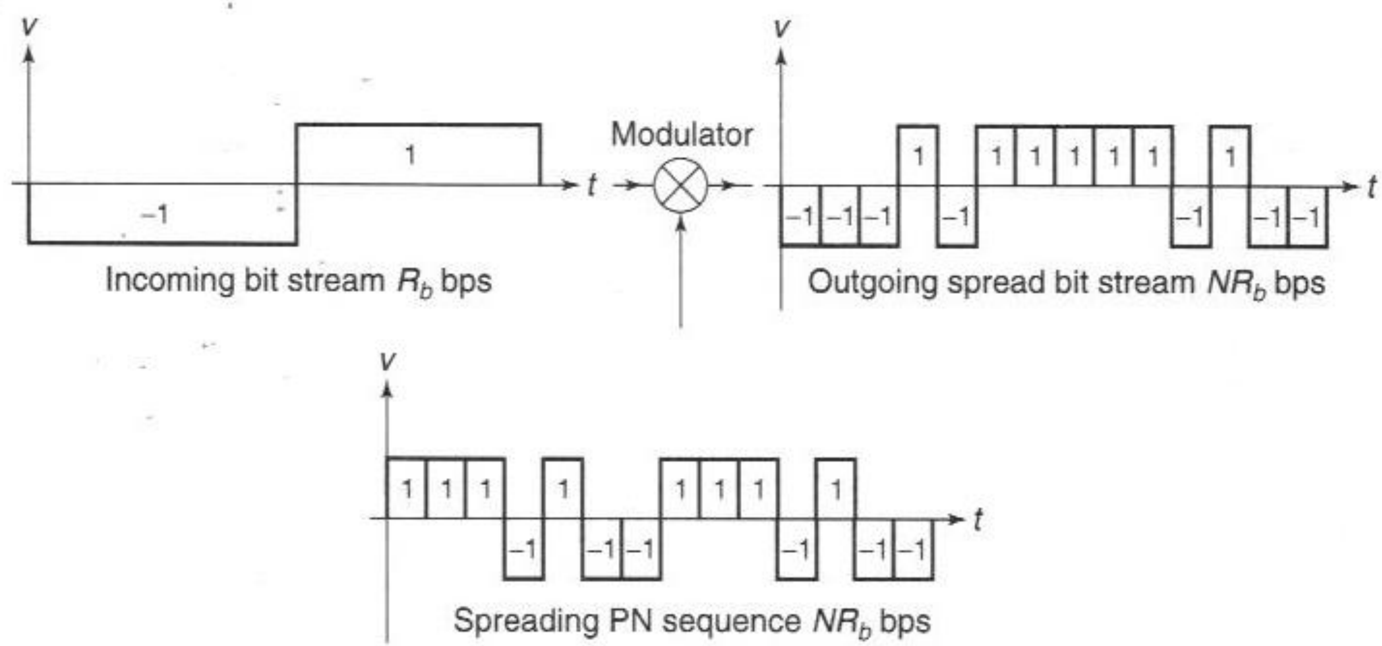


FIGURE 6.16 The basic principle of a direct sequence spread spectrum (CDMA) system. Each incoming message data bit is multiplied by the same PN sequence. In this example the message sequence is −1 +1 and the PN sequence is +1 +1 +1 −1 +1 −1 −1.

Fig. 6.16 from 2nd ed. text

CDMA Codes

- CDMA uses Pseudo Noise (PN) codes
- Codes appear random, have flat spectrum
- Spectrum looks like noise
  - Codes are usually very long
  - PN (m) sequence, Gold, or Kasami codes
  - Not quite orthogonal, but close enough
  - GPS uses Gold codes (1023 chips long)

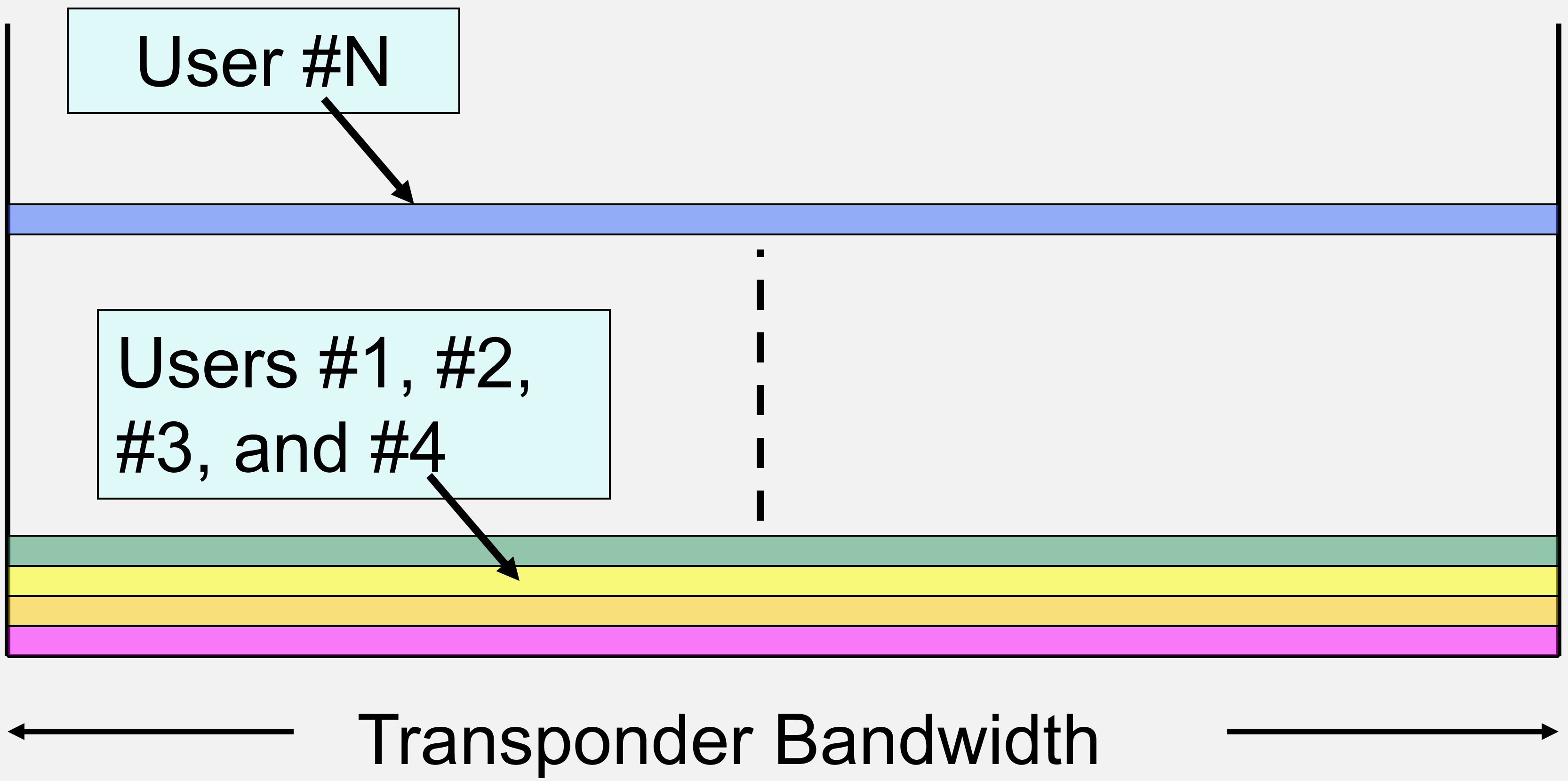
FH-SS CDMA

- *Frequency Hopping Spread Spectrum* (FH-SS)
  - Transmitter changes frequency in random sequence
  - Hops across available bandwidth
  - FHSS is rarely used in satellite links
  - Bluetooth uses FHSS for multiple access

DSSS - CDMA

- *Direct Sequence Spread Spectrum* = DSSS
  - Multiply message by high-speed *chip* sequence
  - Signal occupies full bandwidth all the time
- Signals are overlaid (in time and frequency)
- DSSS is used in some satellite links (e.g. GPS, military communications)

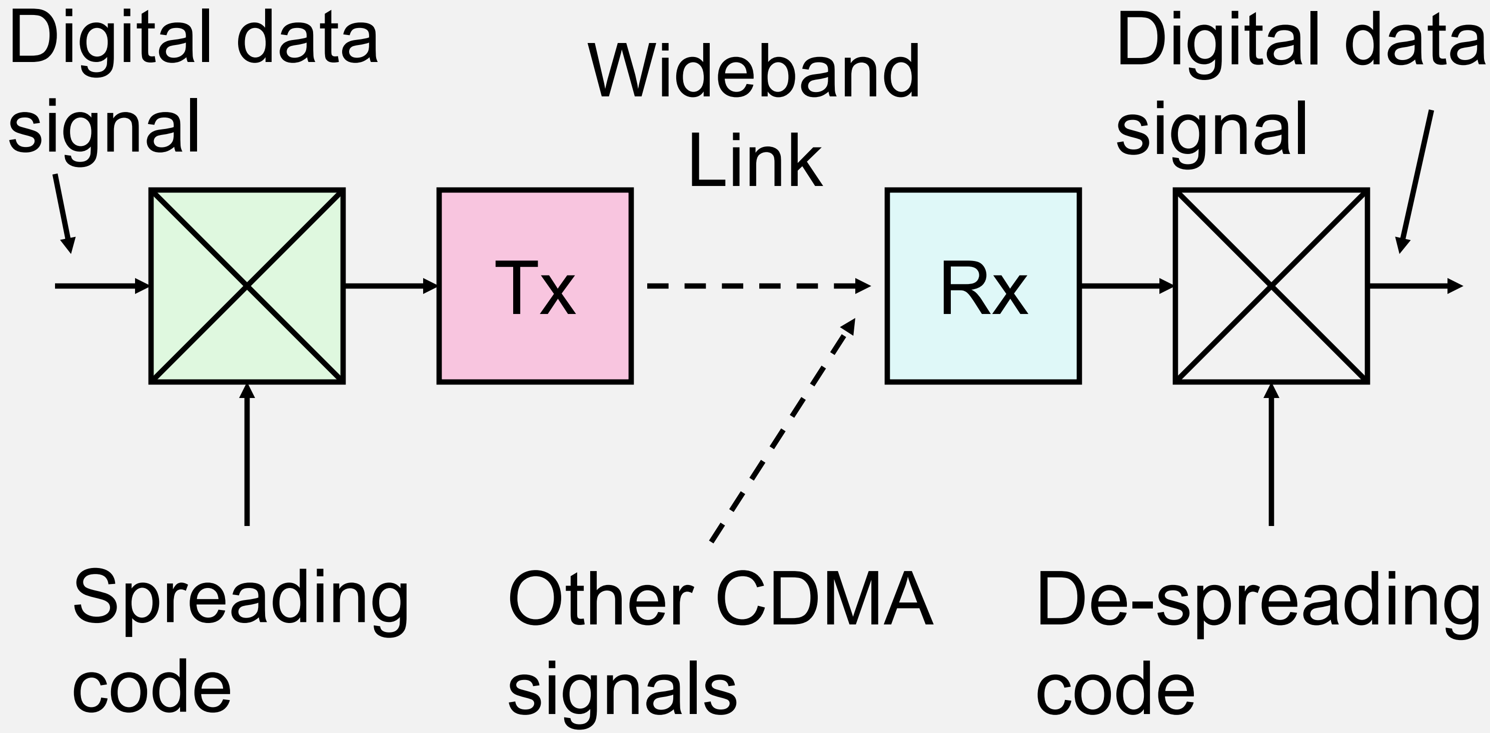
DSSS-CDMA Spectrum



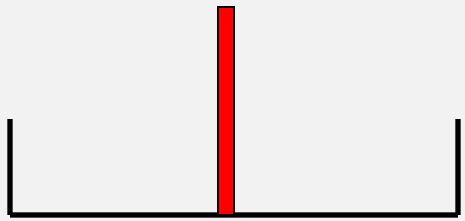
CDMA

- DS-SS signal energy is *spread* across a wide bandwidth
- Signal gets ‘buried’ in the noise of a receiver
- Detection of signal by outside parties is difficult
- Intended receiver has a copy of the chip code
- Correlation of signal with replica code recovers the message bits (*de-spreads* the signal)

Direct Sequence Spread Spectrum Link

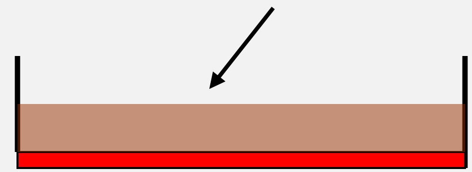


Direct Sequence CDMA



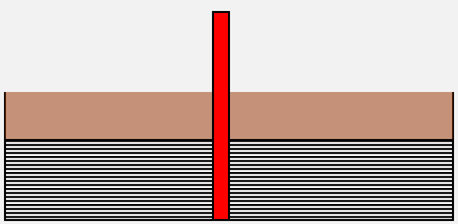
Narrow-band data

Receiver noise



Narrow-band data spread over CDMA bandwidth

Other spread signals added



De-spreading process brings signal out of noise

CDMA - Security

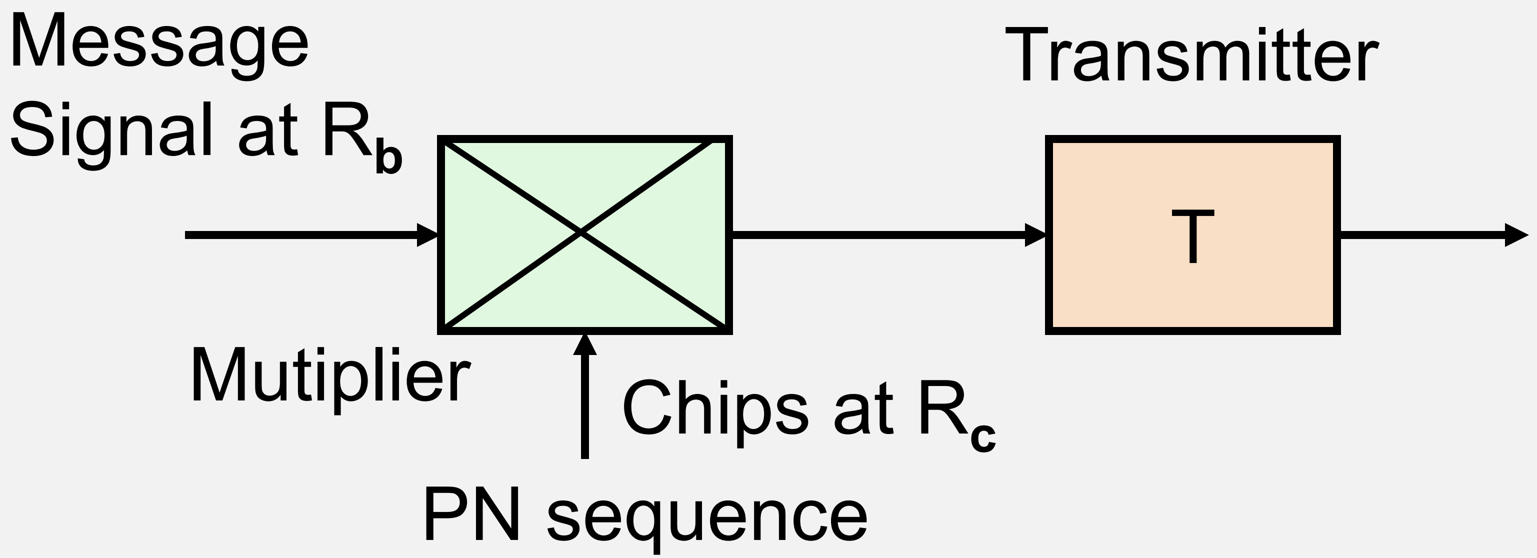
- CDMA was developed by the military to reduce the likelihood of a hostile intercept
- An additional layer of protection can be provided by varying the chipping sequence in time

CDMA - GPS

- CDMA is used with the GPS satellites with each assigned its own PRN code (a Gold code)
- The coarse acquisition (C/A) codes for GPS are 1023 chips long and repeat every 1 ms
- They are modulated by navigation message bits at the rate of 50 bps or 20 ms in time
- The duration of a message bit corresponds to 20 PRN sequences

DS-SS: Transmitting

- Message is multiplied by chip rate sequence ($R_c >> R_b$)



Encoding - Transmitter

- Multiply message by high speed PN bit stream
- Result is a stream of *chips*
- RF energy is spread over a wide bandwidth
- Transmit with same power as message signal

Conversion of a stream of message bits to chip bits

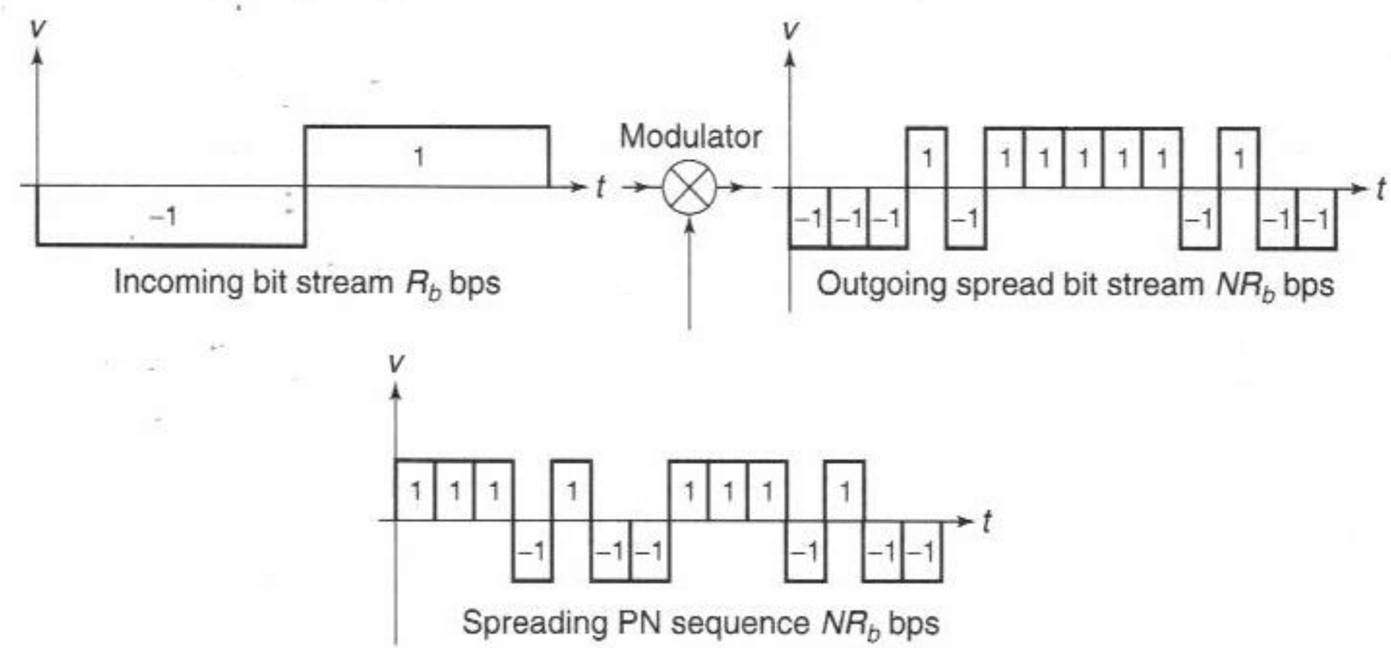
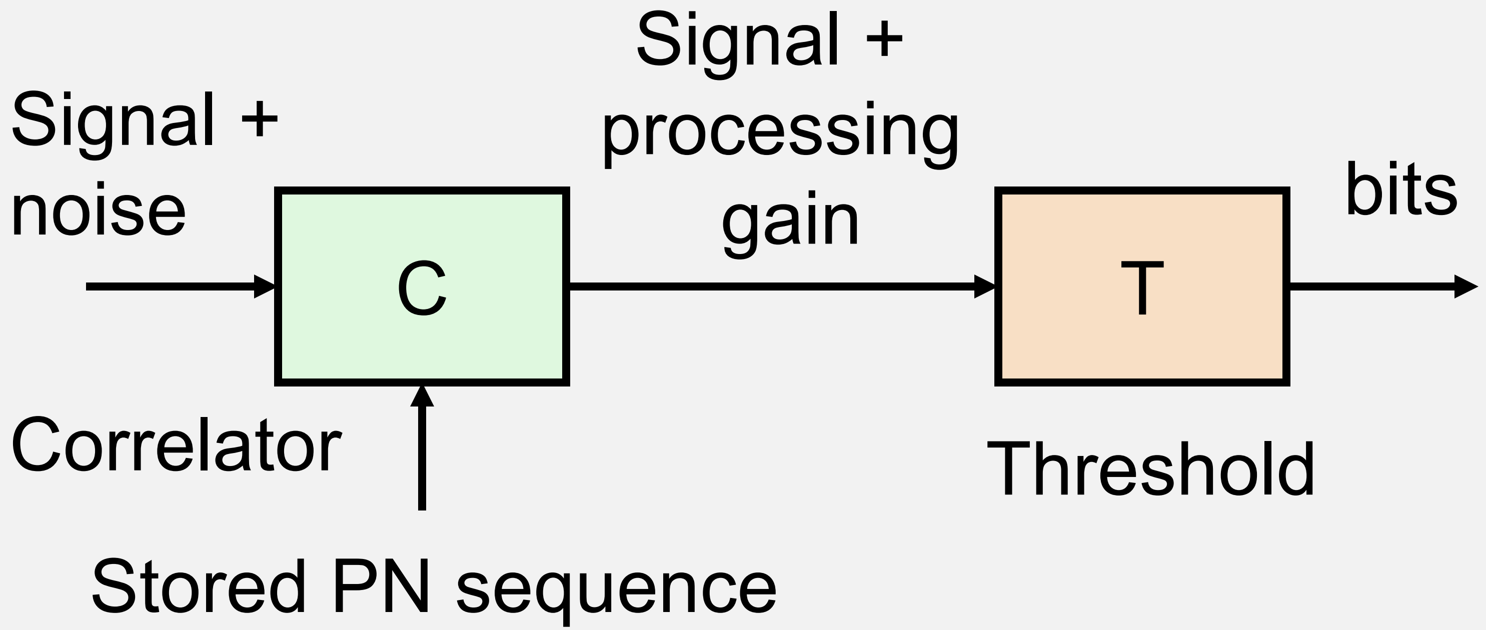


FIGURE 6.16 The basic principle of a direct sequence spread spectrum (CDMA) system. Each incoming message data bit is multiplied by the same PN sequence. In this example the message sequence is −1 +1 and the PN sequence is +1 +1 +1 −1 +1 −1 −1.

Note: for GPS there are 20 chipping sequences for each message bit

DSSS - Receiving

- Received signal is below thermal noise at receiver input
- Correlation of sequences gives *coding gain*

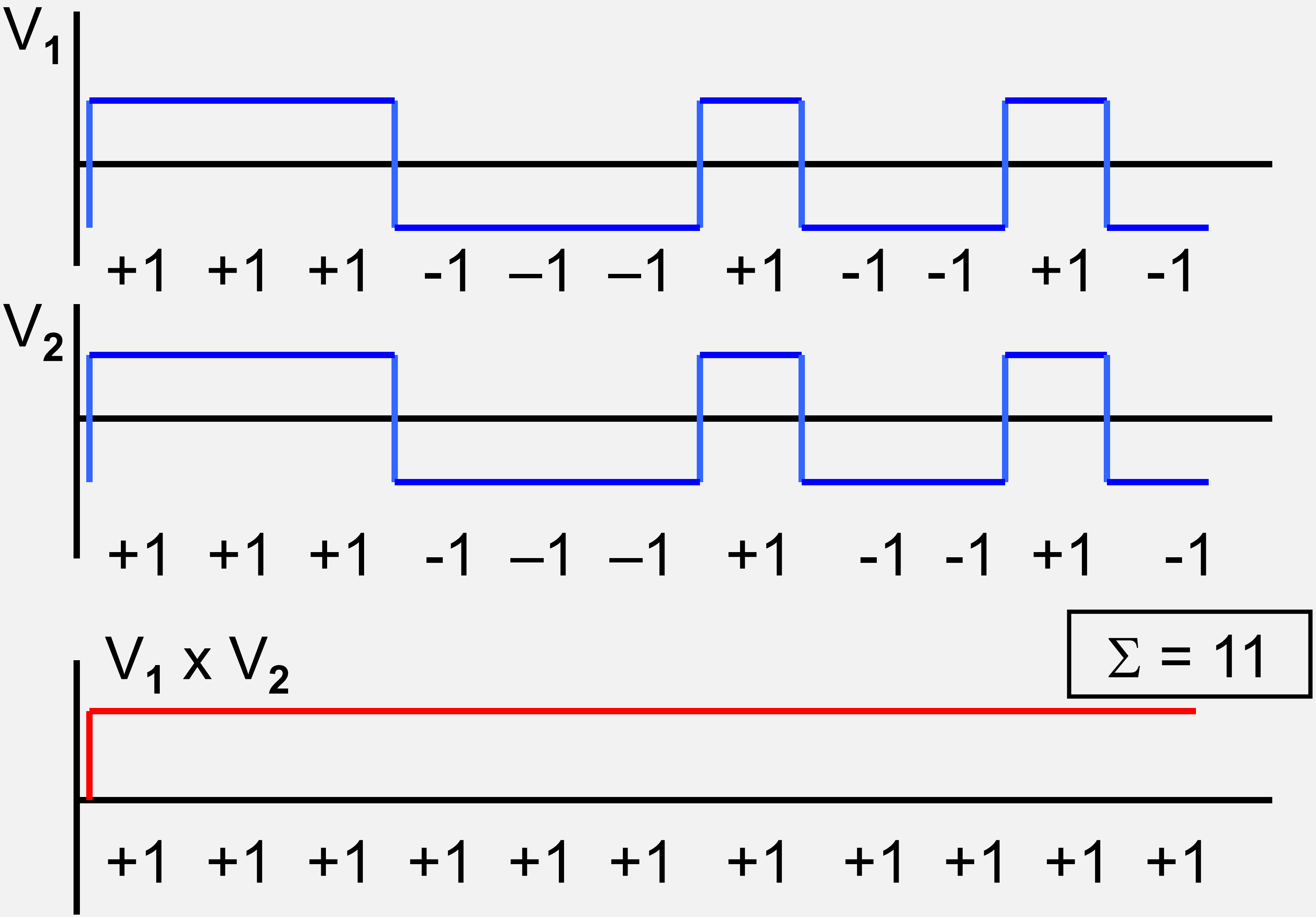


Recovery - Receiver

- In receiver, multiply received chip signal by stored sequence and integrate result
- If sequence is present in signal and properly aligned, the integrator has finite output
- If sequence is not present or not aligned correctly, integrator output tends to zero

DSSS CDMA

- Transmitted signal is chip sequence
- To recover message, we must remove chips
- Receiver must know chip sequence
- Multiplying received signal by chip sequence recovers message signal
- Process is known as *correlation*
- Correlator must be time synchronized to received signal



Baseband correlation example

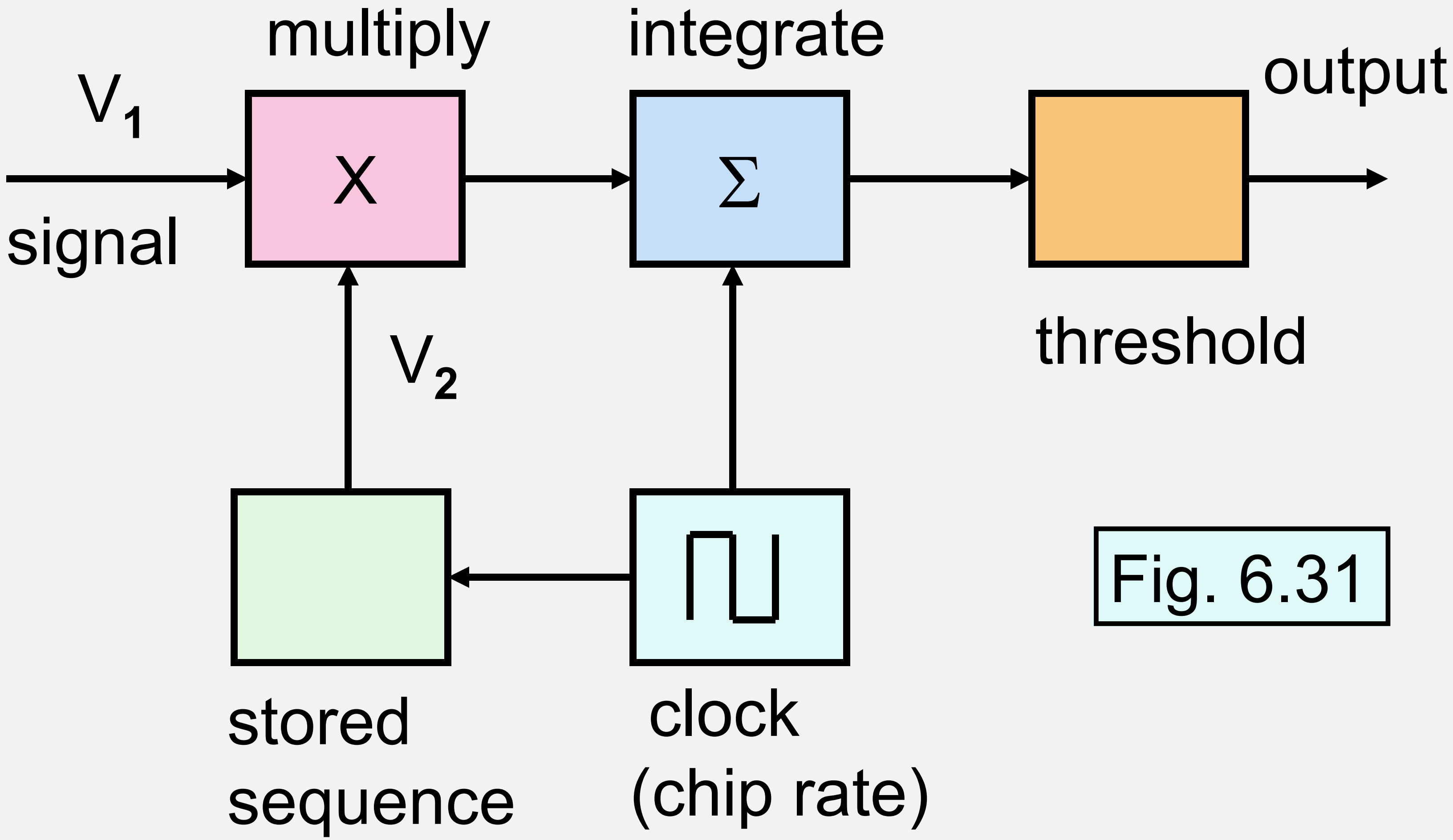


Fig. 6.31

Baseband correlator

+1 +1 +1 -1 -1 -1 +1 -1 -1 +1 -1	V_1
+1 +1 +1 -1 -1 -1 +1 -1 -1 +1 -1	V_2

Multiply

+1 +1 +1 +1 +1 +1 +1 +1 +1 +1 +1	$V_1 x V_2$

Integrate

$\boxed{\Sigma = +11}$ Data bit = 1

Sequence aligned correctly: maximum output

Correlation Example

+1 +1 +1 -1 -1 -1 +1 -1 -1 +1 -1	V_1
-1 -1 -1 +1 +1 +1 -1 +1 +1 -1 +1	V_2

Multiply

-1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1	$V_1 x V_2$

Integrate

$\boxed{\Sigma = -11}$ Data bit = -1

Sequence aligned correctly: minimum output

Correlation Example

+1 +1 +1 -1 -1 -1 +1 -1 -1 +1 -1 +1 V_1

+1 –1 +1 +1 +1 -1 -1 -1 +1 -1 -1 +1 -1 V_2

Multiply

-1 +1 +1 -1 +1 +1 -1 -1 +1 -1 -1 -1

Integrate

$$\boxed{\Sigma = +5 - 6 = -1}$$

Sequence not aligned correctly: 1 bit offset

Correlation Example

$\boxed{+1\ +1\ +1\ -1\ -1\ -1\ +1\ -1\ -1\ +1\ -1}$ +1 +1 V_1

$\boxed{+1\ +1\ +1\ -1\ -1\ -1\ +1\ -1\ -1\ +1\ -1}$ V_2

Multiply

$\boxed{+1\ -1\ -1\ +1\ -1\ +1\ -1\ -1\ +1\ +1\ -1}$

Integrate

$$\boxed{\Sigma = +5 - 6 = -1}$$

Sequence not aligned correctly: 2 bit offset

Correlation Example

+1 +1 +1 -1 -1 -1 +1 -1 -1 +1 -1 | +1 +1 +1 V_1

+1 +1 +1 -1 -1 -1 +1 -1 -1 +1 -1 V_2

Multiply

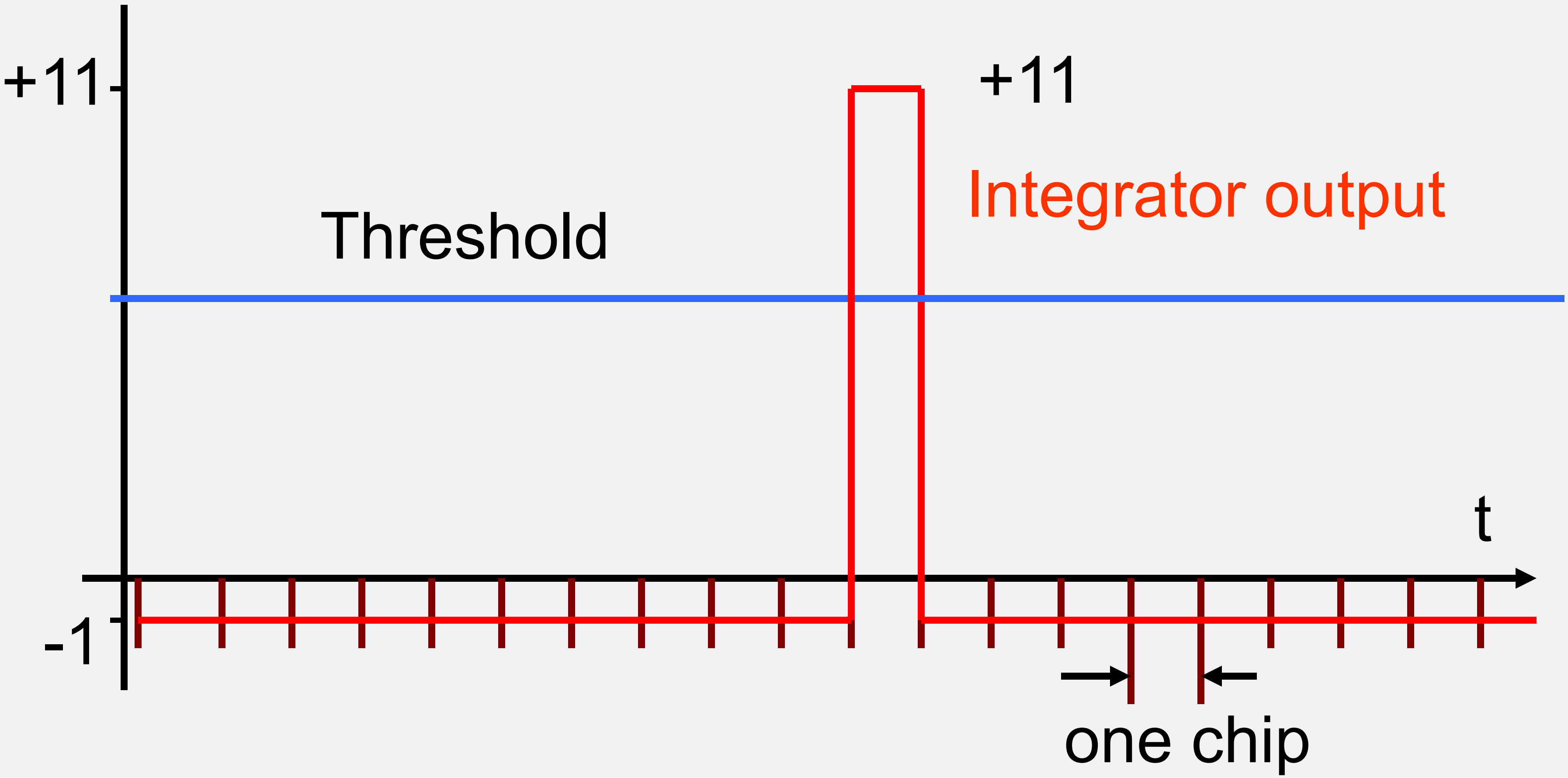
-1 -1 -1 -1 +1 +1 +1 +1 -1 +1 -1

Integrate

$$\boxed{\Sigma = +5 - 6 = -1}$$

Sequence not aligned correctly: 3 bit offset

Correlation Example



Integrator output with 11 bit code and misalignments from –11 to +9 chips - the spreading code has excellent autocorrelation properties

Baseband Correlation

- This example used 11 bit Barker code
- *Barker code* is ideal in the sense that any misalignment results in minimal integrator output (magnitude ≤ 1)
- Barker code has the best sidelobe performance (sidelobes have amplitude 1/N that of the peak)
- Only a few Barker codes are known (max: 23 bits)

Baseband Correlation

- M-sequence and Gold codes do not have ideal cross-correlation
- Selection of code set is required to keep cross-correlation time sidelobes low
- Certain offsets between codes will produce outputs greater than one
- But must still be low compared to the chip count, *M*

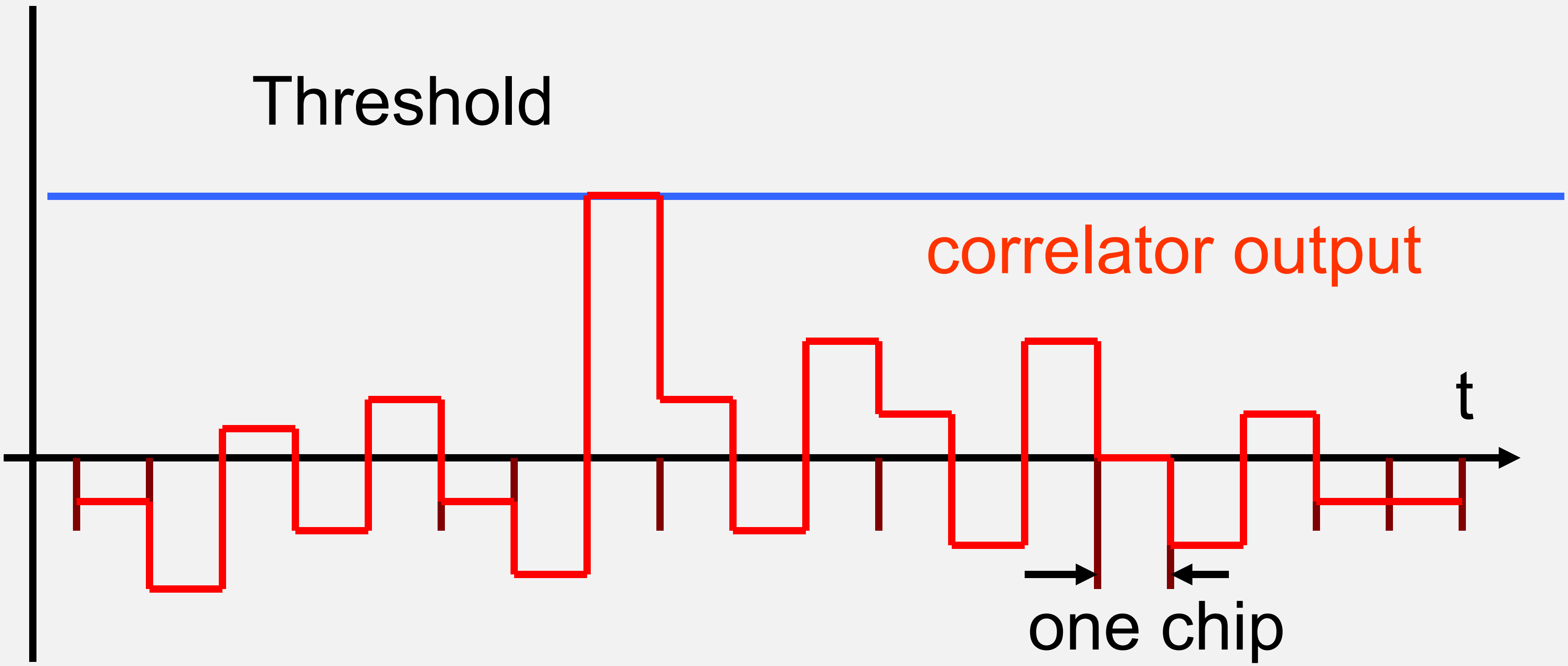


Illustration of cross-correlation with a non-ideal 11 bit code over 19 chip alignments

Correlation

- To obtain output from correlator, stored code must be aligned correctly with the input code
- Correlator output then maximizes in magnitude and triggers detection of data bit (+1 or -1)
- The correlator is said to *despread* the CDMA signal
- Signal bandwidth is greatly reduced
- Timing resolution is half of one chip or better

Baseband Correlator for Despreading CDMA signals

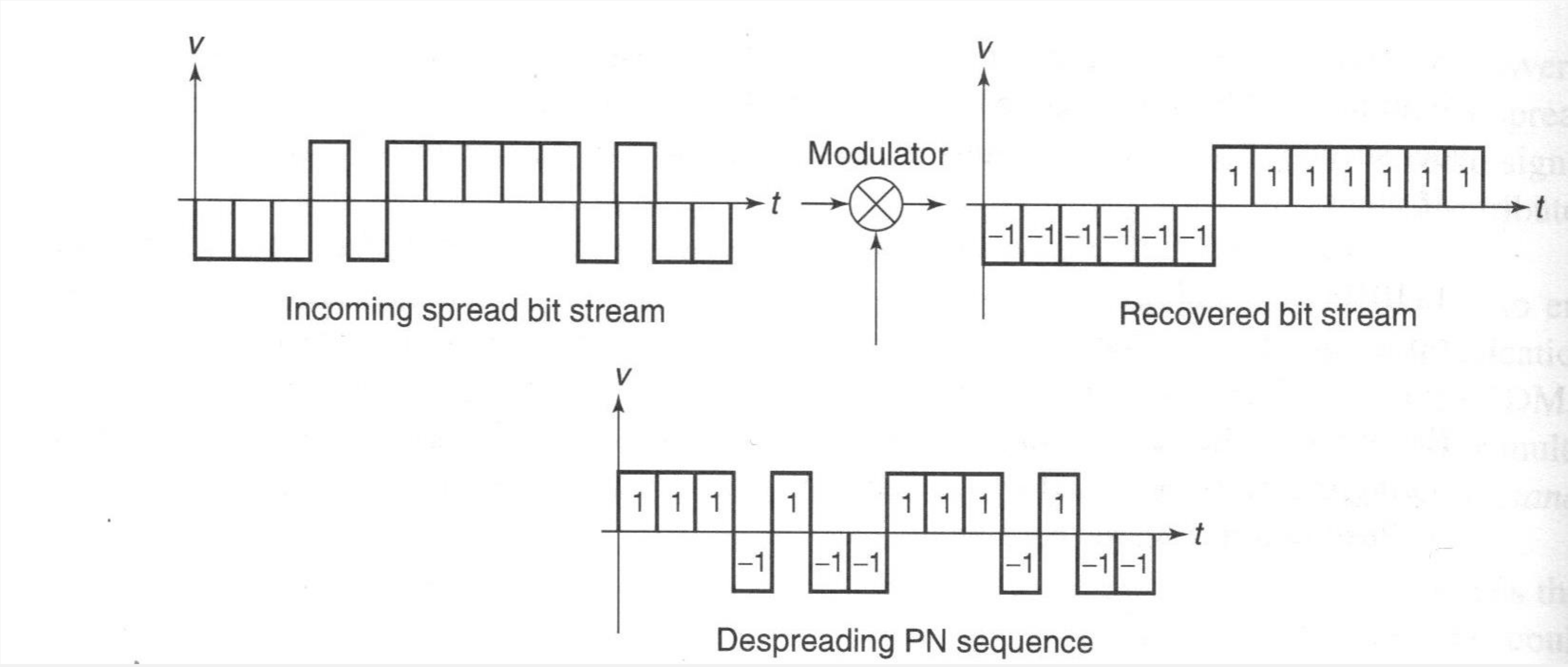


Fig. 6.18 from 2[nd] ed. text

Detection of CDMA signal

- Other CDMA signals may be present at the correlator input
- Ideally their correlation with the PN sequence is +1 or -1, i.e., very low compared to the PN chip count, M
- Given a large value of M, the bits can be detected against a ‘noisy’ background of CDMA signals

Detection of CDMA signal

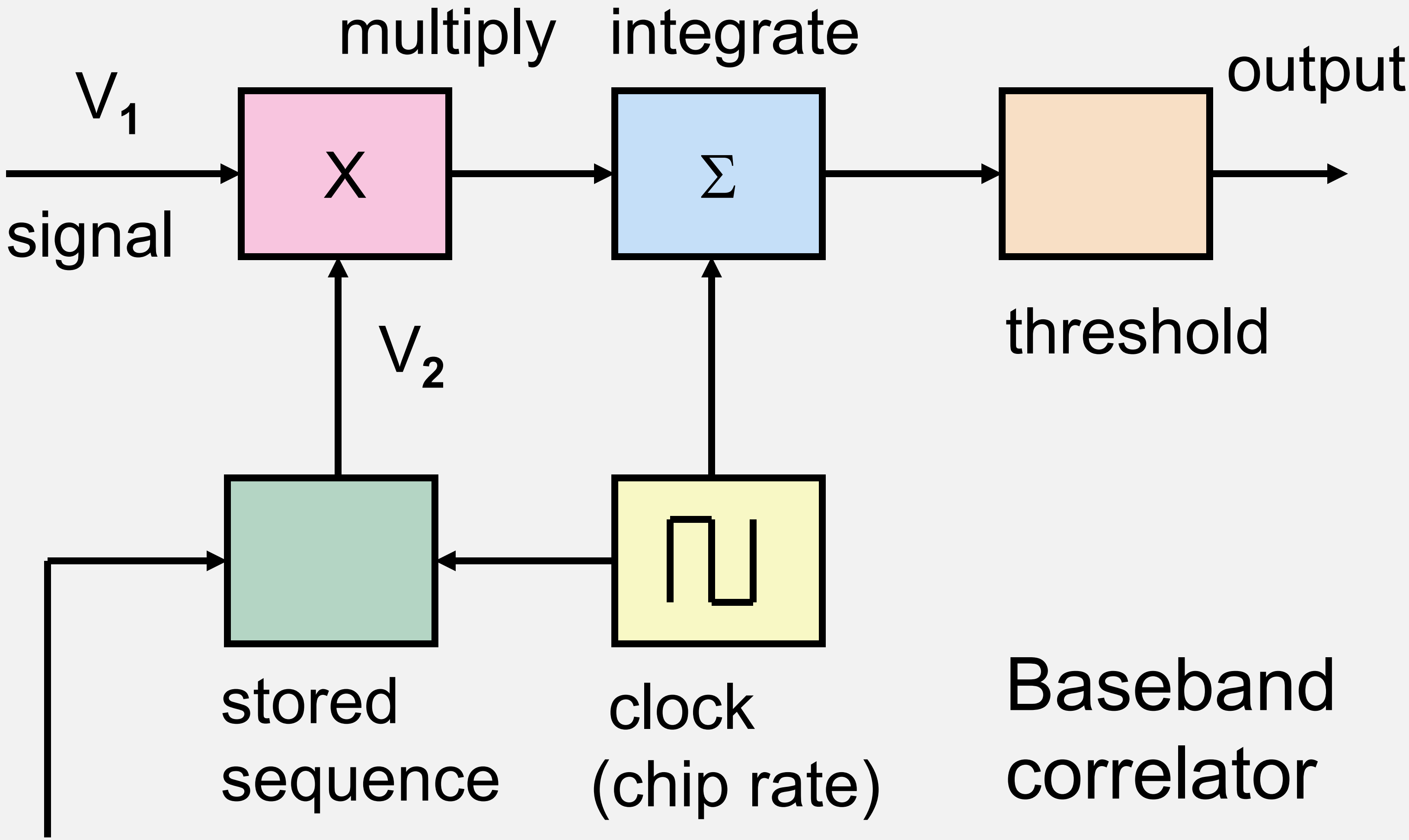
- Good PN codes have low correlation when misaligned
- Barker codes are perfect (up to 23 chips)
- GPS uses less-than-perfect 1023 chip Gold codes

Acquiring Synchronization

- *Spread spectrum* signals are typically below noise
- Receiver cannot detect signal with conventional demodulator since C/N << 0 dB
- Must search for correct code timing by trial and error
- When code timing is correct, correlator output goes above threshold and a bit value is read

Acquiring Synchronization

- When all possible start times have been tried, one should give an output
- If there is no output:
  - signal is absent
  - signal is too weak to cross threshold
  - wrong code loaded into sequence generator



Baseband correlator

Set start point in sequence

IF Correlation

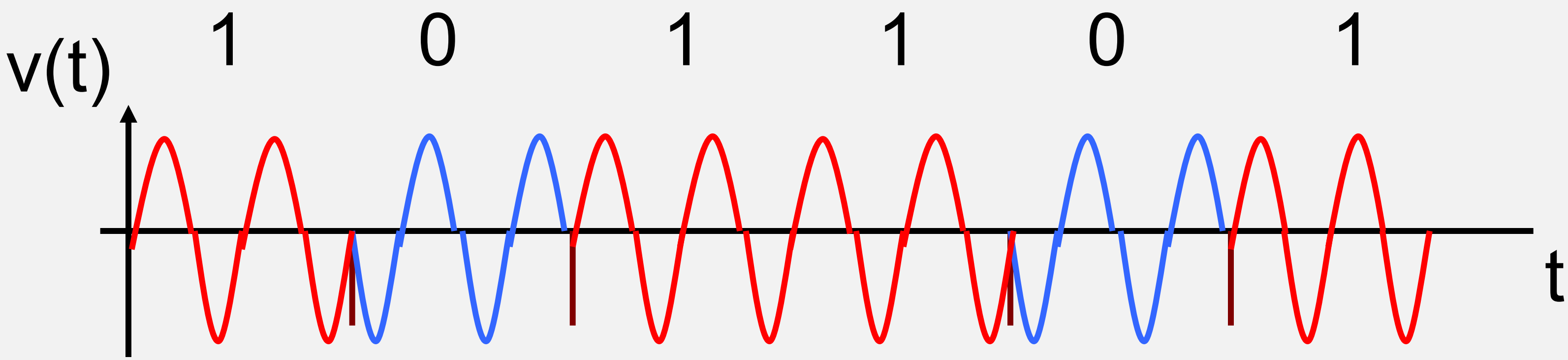
- To this point for simplicity we have analyzed CDMA correlation at baseband
- Real systems will employ RF for transmission and IF for signal processing
- Correlator can operate on signals at IF

IF Correlation

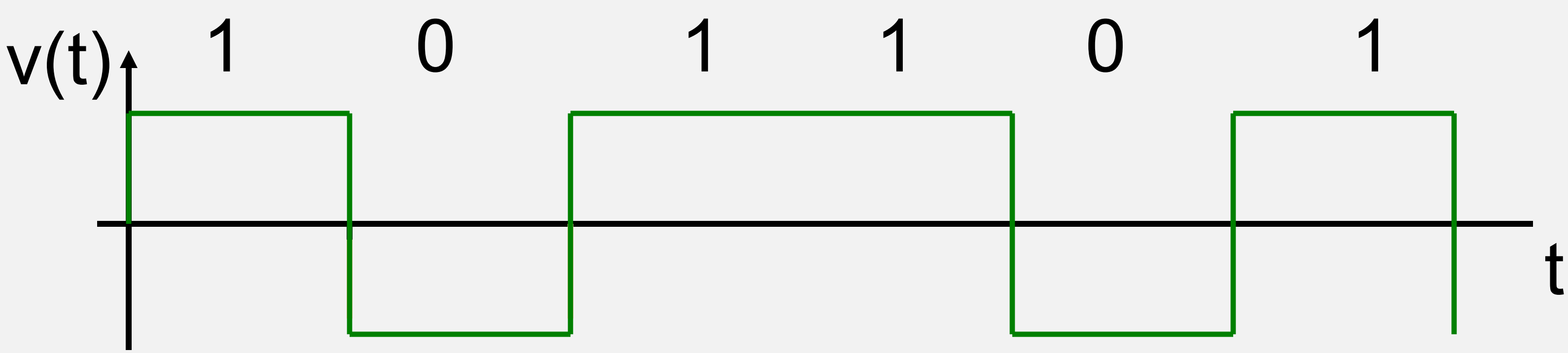
- Practical CDMA systems use a sequence generator and a BPSK modulator
- Same principle as baseband correlator, except signals are now BPSK at IF
- PN sequences are BPSK waveforms with 0˚ and 180˚ phase shifts
- Coding chip information is now carried in the phase of the IF signal

Code timing correct

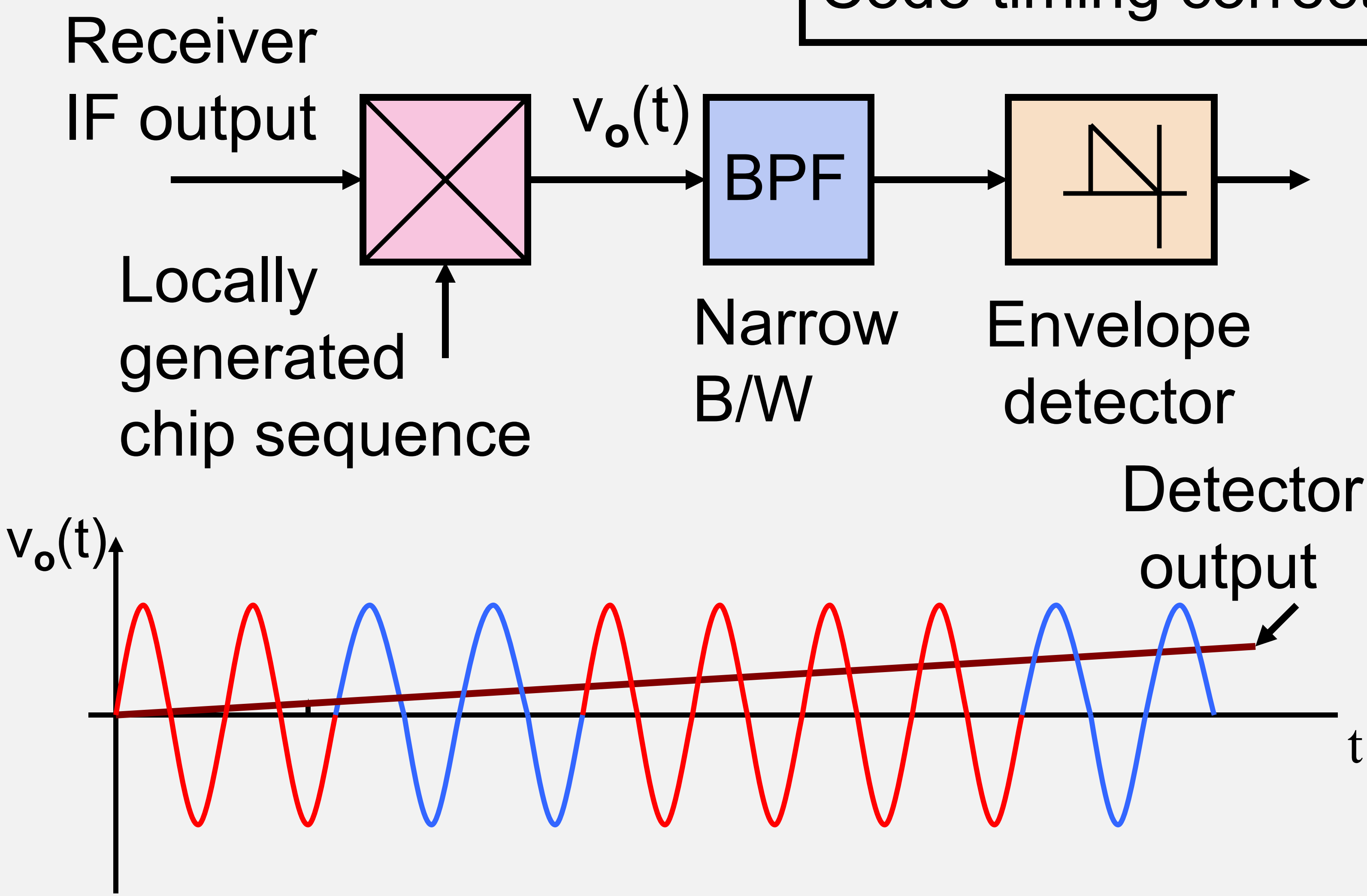
Received IF signal with chipping (noise omitted)

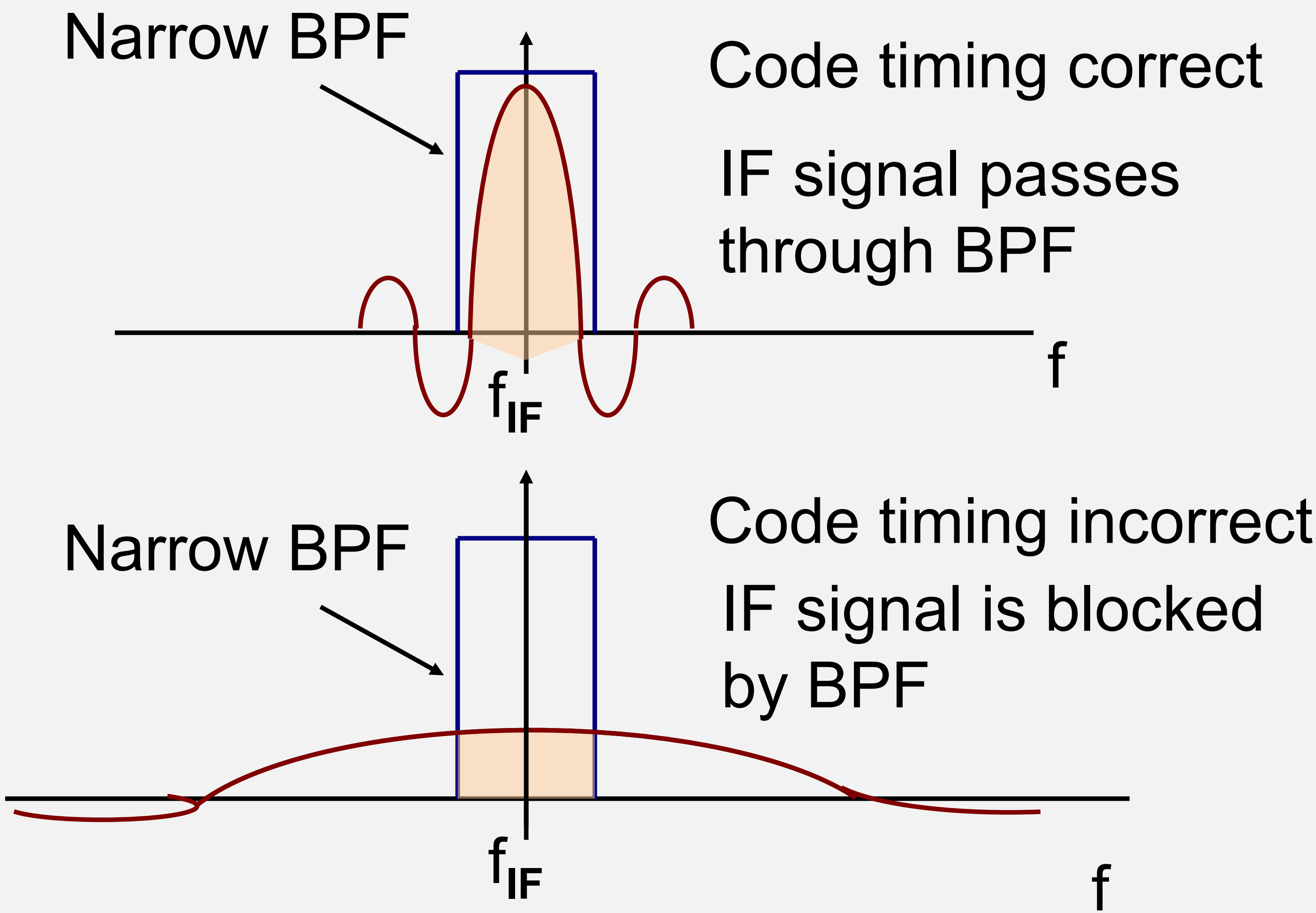


Locally generated code signal



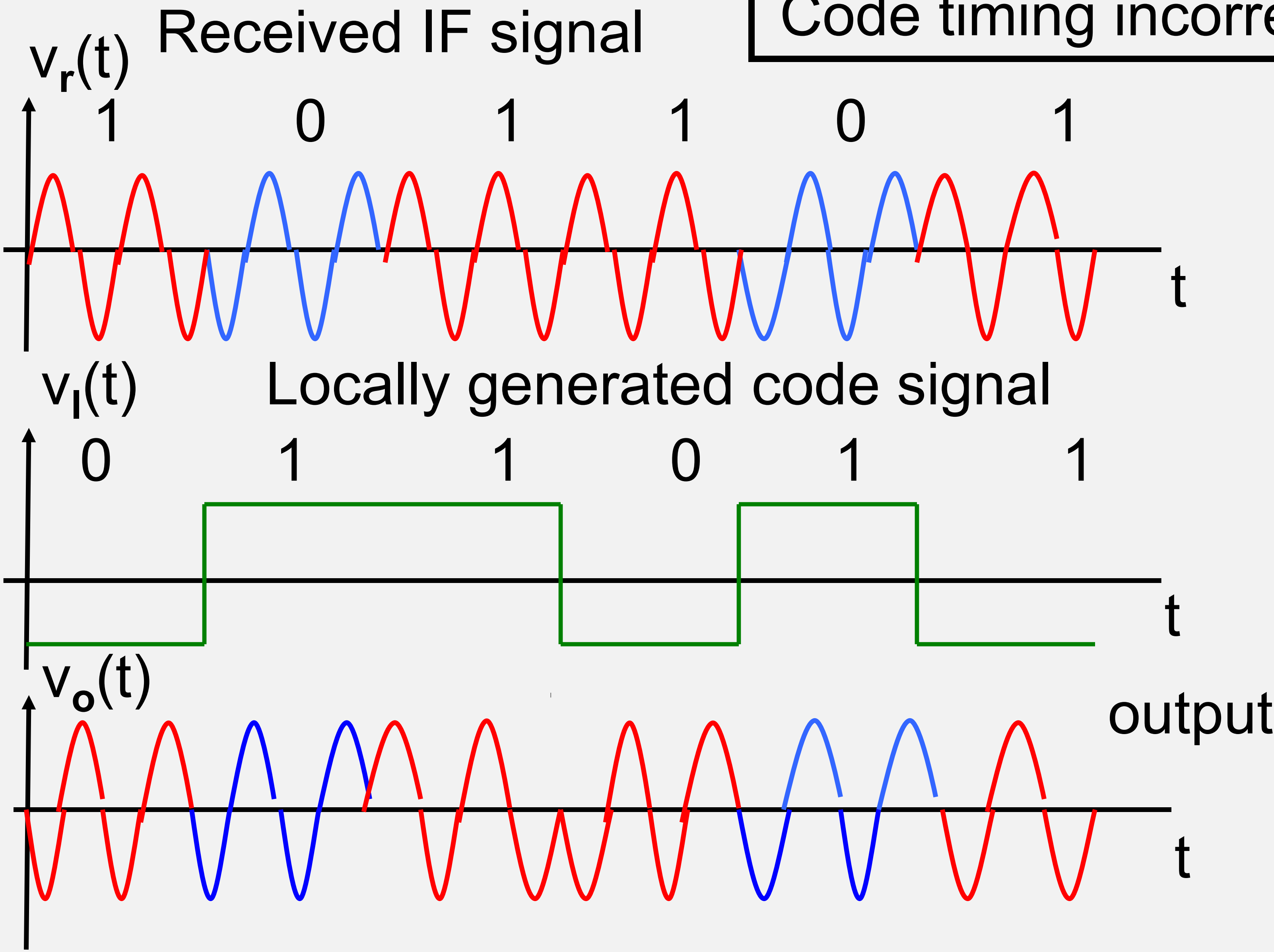
Code timing correct





Spectra at multiplier output

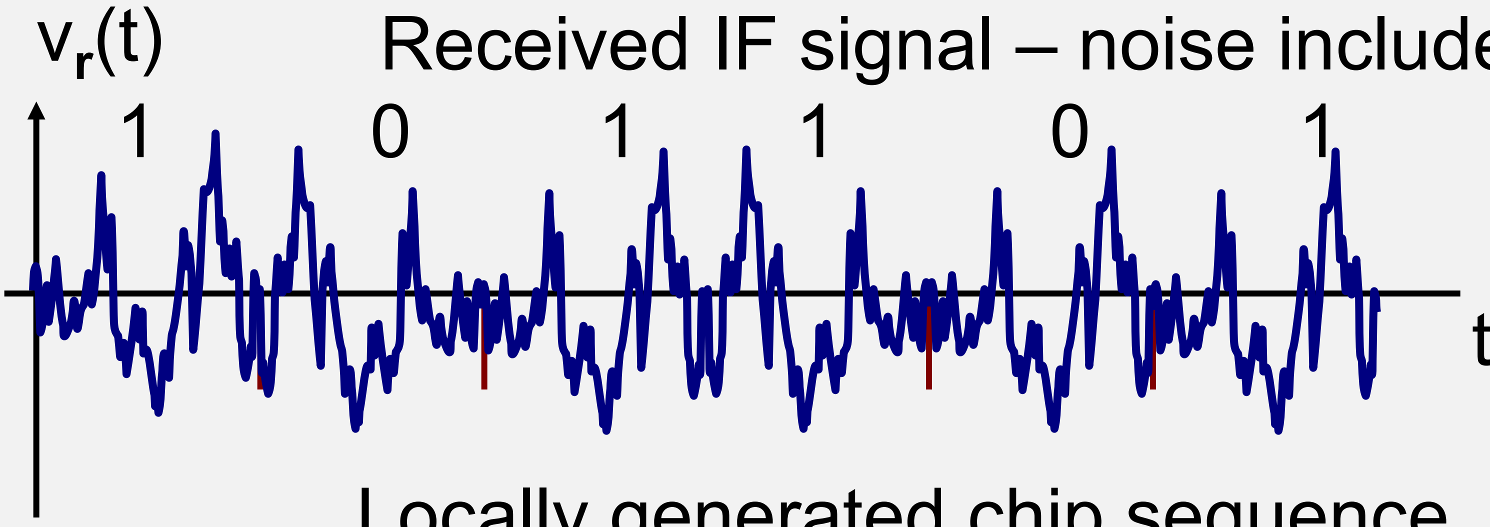
Code timing incorrect



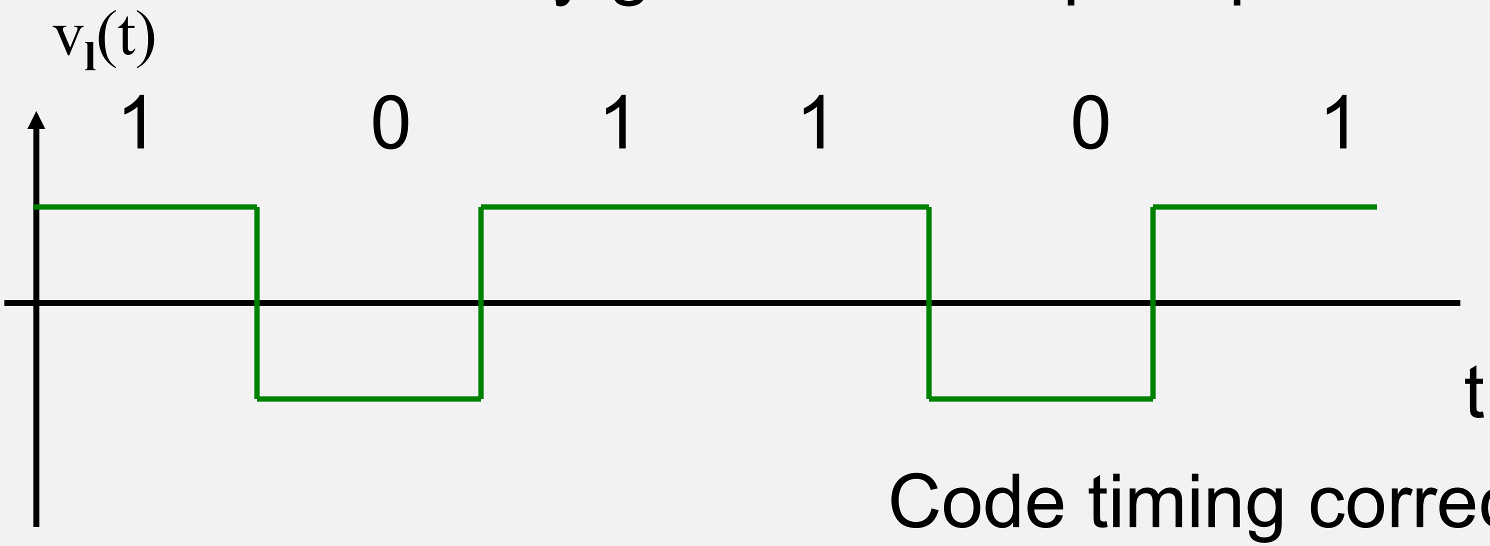
IF correlator

- We apply in receiver a chip sequence that matches the code sequence received
- When we have an exact match in time and phase transitions then all chips are removed from signal
- IF signal can then pass through narrow bandpass filter – the signal is despread
- Filter bandwidth = bandwidth of message signal

Received IF signal – noise included

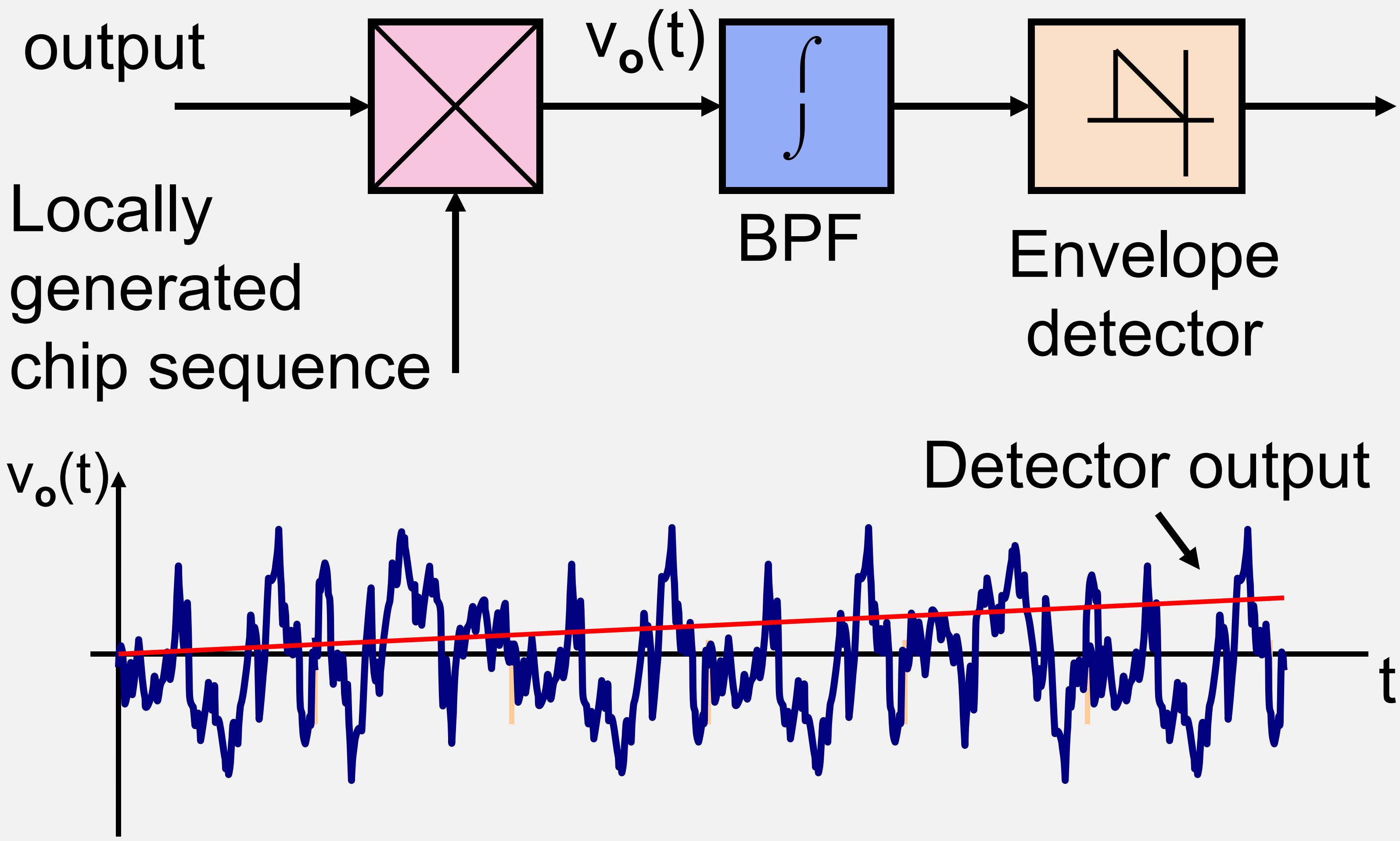


Locally generated chip sequence



Code timing correct

Signal detector



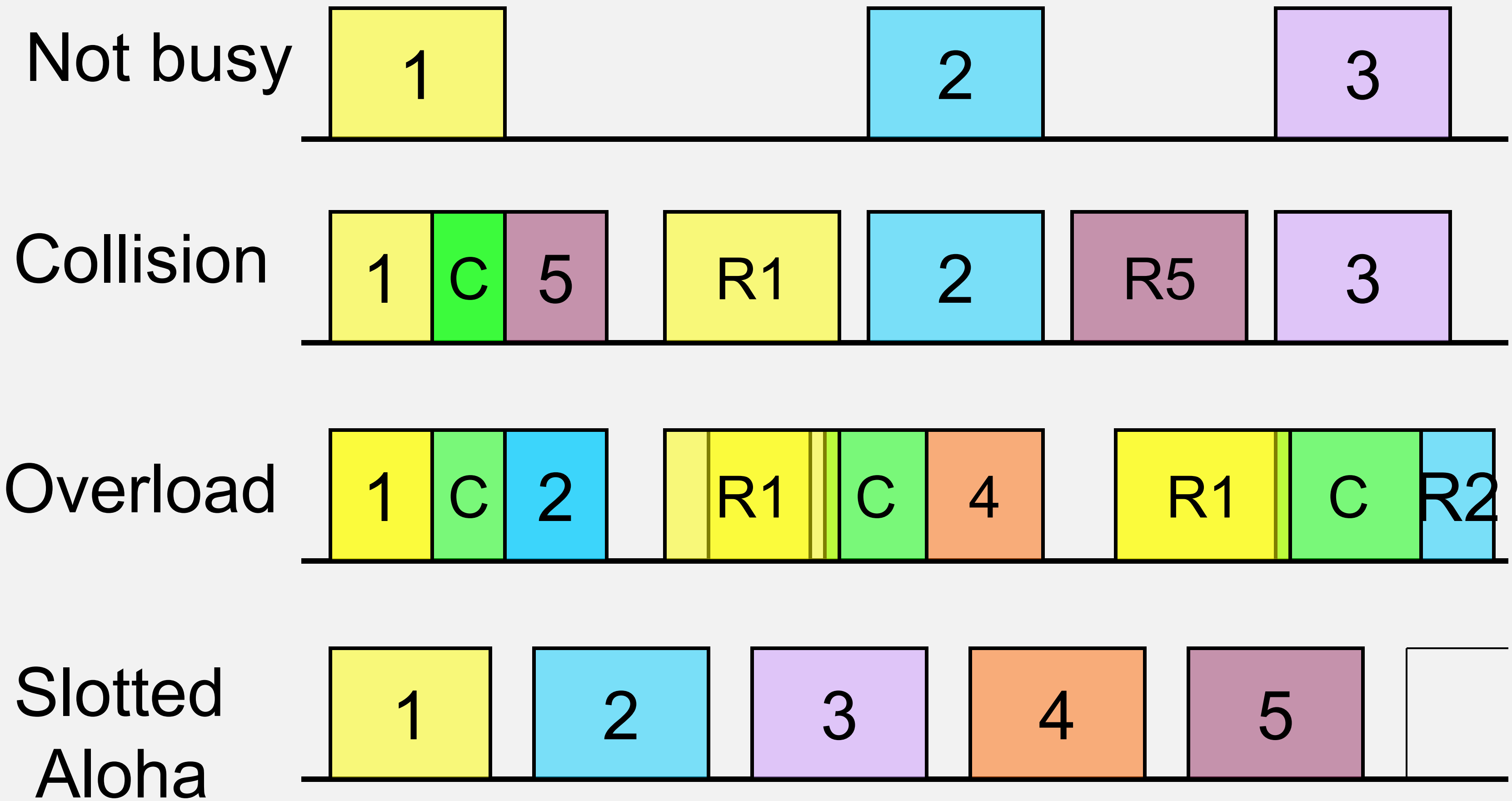
CDMA Applications

- Military
  - Anti-Jam (AJ)
  - Low Probability of Intercept (LPI)
- Commercial
  - Globalstar LEO SatCom system
  - GPS
  - Digital Cellular Radio Systems

Random Access - RA

- Uncontrolled FDMA
- Transmit in the allocated spectrum
- Send packet at a random time, okay if traffic density is low
- Collisions are detected at transmitter
  – Re-transmit with a random delay

Random Access

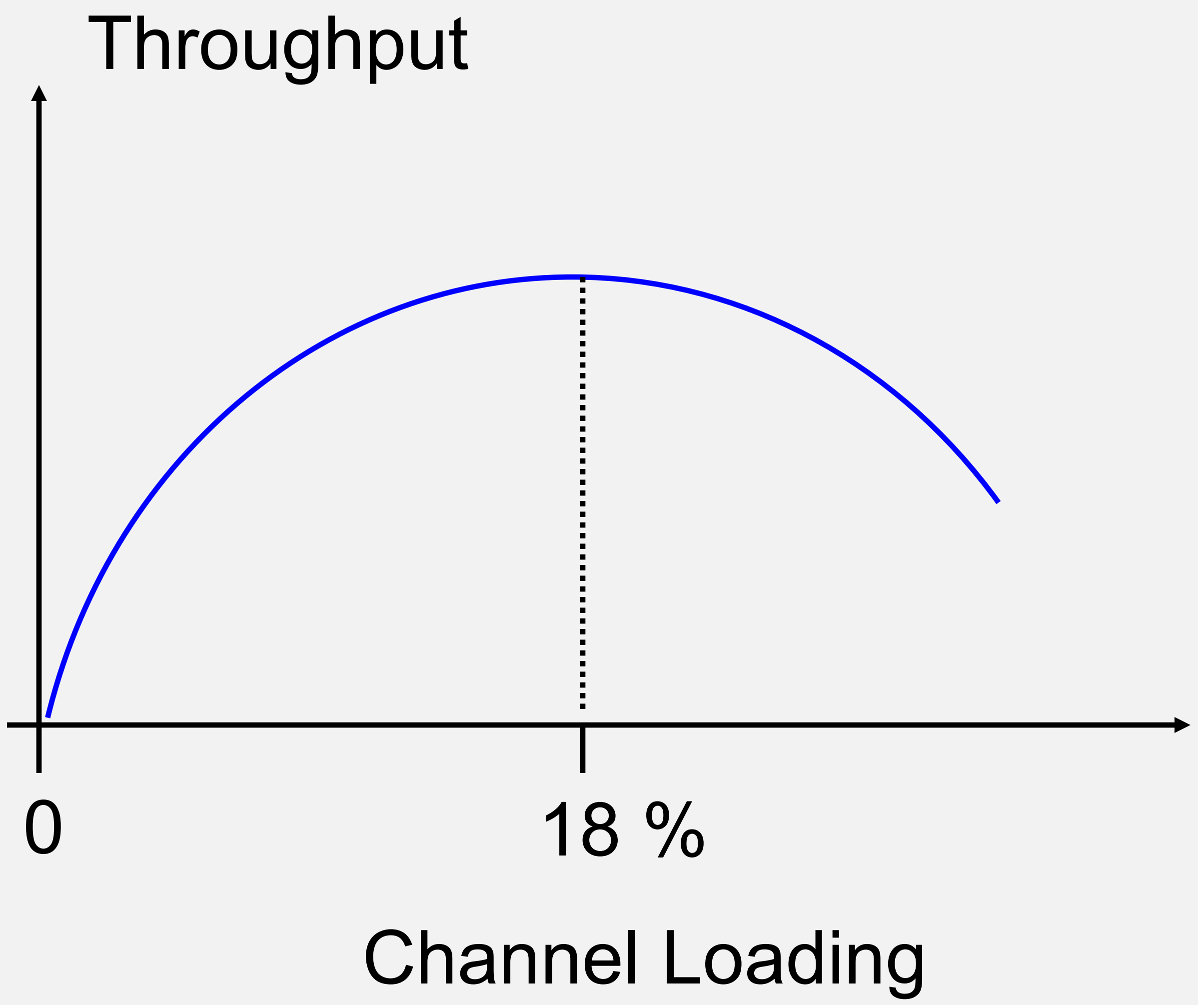


C = Collision R = Repeat

Random Access - RA

- True RA has maximum loading of 18%
- Each packet is re-transmitted 2.7 times on average with 18% loading
- Any increase in loading above 18% reduces throughput so much that system collapses
- Most suitable when traffic bursts are short and intermittent (e.g., internet access – outbound)

Random Access Performance



Random Access

- Random access techniques were developed at University of Hawaii
- Whence the name ‘Aloha’
- An improvement to the original Aloha protocol uses discrete time slots for transmission
- Loading for slotted Aloha RA can reach 36% and higher

Collision Detection

- Random Access requires collision detection
- Data must be sent in packets with error detection capability
- Checksum or Cyclic Redundancy Check (CRC) used to detect errors in packet
- CRC is usual on satellite links

Packet Transmission

Typical packet structure - AX.25 protocol for amateur radio satellite system (after Fig. 6.29)

Flag	Add	Con	Message data	CRC

Flag = Unique word to signal start of packet (01111110)

Add = Address(es) – call signs

Con = Control

CRC = Cyclic Redundancy Check

Demand Assignment (DA)

- With *Fixed Access* satellite, capacity for each Earth station is allocated in advance
- With *Demand Access* a communication channel is assigned upon request
- Channel is unassigned when no longer needed
- Works when a link does not need to be constantly maintained
- Makes more efficient use of resources

Demand Assignment Multiple Access (DAMA)

- *Demand Assignment Multiple Access* (DAMA) manages DA amongst multiple clients
- Examples of DAMA include cellular phone service and VSAT networks
- Packet techniques are usually applied
- Two different channels are required:

1) CSC – *Common Signaling Channel*
2) Communication Channel

DAMA

- User sends request via the CSC to the controller earth station, also called *gateway* earth station
- The controller assigns a pair of channels for sender communications (outbound, inbound)
- Inbound to gateway could be FDMA in one transponder while outbound from gateway is TDMA in another transponder
- Internet service will usually be asymmetric in its outbound / inbound channels owing to differing load requirements

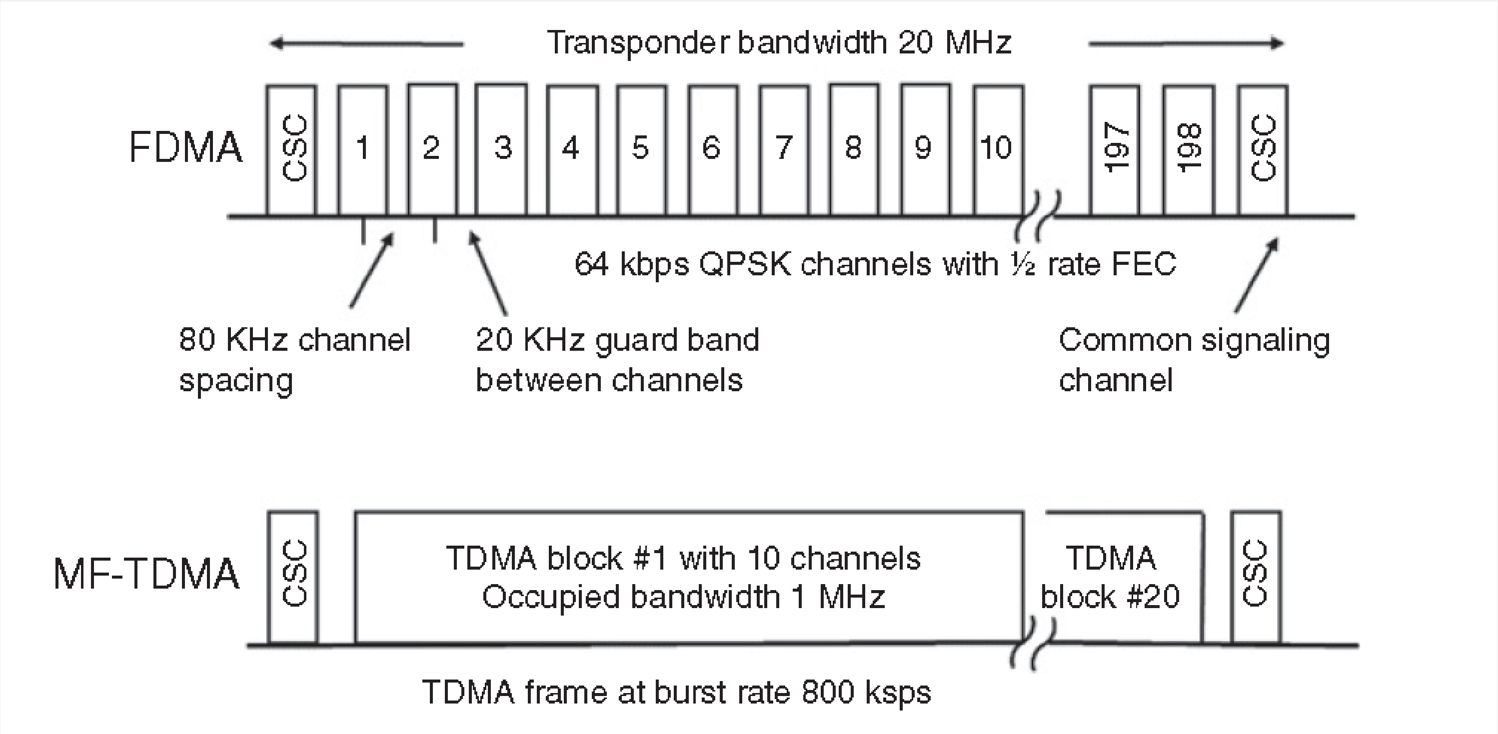


Fig. 6.26: FDMA and MF-TDMA loading of 20 MHz transponders, each with 200 channels. Note the two channels for common signaling.

DAMA

- Rapid response to a CSC demand is essential
- Required: seamless link, available on demand with good BER, and no dropouts in data throughput
- Throughput will typically vary with load on the system, may be time-of-day dependent
- See course text Section 6.10 for detailed discussion

Summary of Multiple Access (1/4)

- FDMA is widely used in satellite links
- Best access technique for uplink of low EIRP terminals
- TDMA is best outbound technique in STAR networks
- TDMA has greater flexibility than FDMA
- Requires burst transmission, higher EIRP than FDMA

Summary of Multiple Access (2/4)

- CDMA is used in military satellite links
- Utilizes pseudonoise (PN) chip sequences
- Leads to *Direct Sequence Spread Spectrum* - DSSS for satcomm
- Provides *low probability of intercept* (LPI) and inherent security
- CDMA receiver must synchronize to CDMA signal below noise level

Summary of Multiple Access (3/4)

- DSSS-CDMA is used in GPS for timing accuracy
- CDMA receiver is more complex than for FDMA or TDMA receiver
- CDMA stations do not need to coordinate their frequency (as in FDMA) or their time of transmission (as in TDMA)

Summary of Multiple Access (4/4)

- RA is simplest of MA techniques in conception
- Widely used with low traffic density systems, e.g., satellite mobile phones, VSATs
- Applied to common signaling channels (CSCs) with DAMA
- Obvious limits to overall throughput
- Requires error detection and retransmission of degraded packets